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Application of first- and second-order adjoint methods to glacial isostatic adjustment incorporating rotational feedbacks

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SUMMARY

This paper revisits and extends the adjoint theory for glacial isostatic adjustment (GIA) of Crawford *et al.* (2018). Rotational feedbacks are now incorporated, and the application of the second-order adjoint method is described for the first time. The first-order adjoint method provides an efficient means for computing sensitivity kernels for a chosen objective functional, while the second-order adjoint method provides second-derivative information in the form of Hessian kernels. These latter kernels are required by efficient Newton-type optimization schemes and within methods for quantifying uncertainty for non-linear inverse problems. Most importantly, the entire theory has been reformulated so as to simplify its implementation by others within the GIA community. In particular, the rate-formulation for the GIA forward problem introduced by Crawford *et al.* (2018) has been replaced with the conventional equations for modelling GIA in laterally heterogeneous earth models. The implementation of the first- and second-order adjoint problems should be relatively easy within both existing and new GIA codes, with only the inclusions of more general force terms being required.

Key words: Sea level change; Inverse theory; Numerical Modelling.

1 INTRODUCTION

In broad terms, the glacial isostatic adjustment (GIA) inverse problem uses palaeo sea level records and related observations (e.g. present-day geodetic velocities and gravity anomalies) to quantitatively reconstruct ice sheet history back into the last glacial whilst simultaneously estimating mantle viscosity variations (e.g. Peltier 1976, 2004; Lambeck *et al.* 1998; Peltier & Drummond 2008; Argus *et al.* 2012; Nakada *et al.* 2015; Roy & Peltier 2018; Li *et al.* 2020). Work on this problem has relied largely on guided forward modelling, but such an approach is feasible only if mantle viscosity is assumed to vary as function of depth alone. The importance of lateral viscosity variations for GIA has now been clearly established (e.g. Latychev *et al.* 2005a; Austermann *et al.* 2013; van der Wal *et al.* 2015; Lau *et al.* 2018; Pan *et al.* 2022), though their inclusion within GIA modelling raises the computational cost dramatically (e.g. Zhong *et al.* 2003; Latychev *et al.* 2005b; Steffen *et al.* 2006; Aa et al., 2013). Experience gained within other fields concerned with large-scale inverse problems (e.g. Lions 1970; Biegler *et al.* 2003; Tröltzsch 2005; Biros & Ghattas 2005a, b; Fichtner *et al.* 2006; Wunsch 2006; Tape *et al.* 2007; Métivier *et al.* 2013; Bozdağ *et al.* 2016) suggests that the application of adjoint methods to GIA will be essential in making progress with the inverse problem while accounting for lateral viscosity variations. Adjoint methods provide an optimally efficient means for determining the gradients within iterative optimization schemes (e.g. Nocedal & Wright 2006), and they also form an essential component within the application of Bayesian methods to large-scale non-linear inverse problems (e.g. Stuart 2010; Bui-Thanh *et al.* 2013; Petra *et al.* 2014; Papadimitriou & Papadimitriou 2015). The development of surrogate models for GIA in laterally heterogeneous earth models (e.g. Love *et al.* 2023) may change this outlook, but their viability is yet to be firmly established.

Theoretical aspects of the adjoint method in the context of GIA have been discussed in a number of papers (Al-Attar & Tromp 2014; Martinec *et al.* 2015; Crawford *et al.* 2018) while others have focused on initial applications (Kim *et al.* 2022; Lloyd *et al.* 2024). The aim of this work is to revisit and extend the approach of Crawford *et al.* (2018) which is the most complete version of the adjoint theory to date. The first point we address is the incorporation of rotational feedbacks into the adjoint problem. The next is the development of second-order adjoint theory. First-order adjoint theory allows for efficient calculation of the derivatives of a chosen scalar-valued objective functional, with these derivatives being required within gradient-based optimization schemes. Similarly, the second-order adjoint theory allows for the action of the Hessian operator (i.e. the second derivative of the objective functional) on a given model perturbation to be efficiently determined. Such calculations are required in the application of Newton-type optimization schemes that offer superior convergence properties (e.g. Nocedal &

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Wright 2006; Métivier *et al.* 2013). Hessian calculations are also necessary within a range of methods for quantifying uncertainties within large-scale and non-linear inverse problems (e.g. Wang *et al.* 1992; Fichtner & Trampert 2011; Bui-Thanh *et al.* 2013; Petra *et al.* 2014; Papadimitriou & Papadimitriou 2015).

Beyond the extensions just mentioned, this paper presents a complete reformulation of the adjoint theory of Crawford *et al.* (2018) with the aim of making these methods more readily understandable and applicable within the GIA community. Crawford *et al.* (2018) built on the rate-formulation for viscoelastic loading developed in Al-Attar & Tromp (2014). An appealing feature of the rate-formulation is that the GIA forward problem can be written as a coupled system of non-linear evolution equations without need for the usual iterative solution of the sea level equation. The rate-formulation also offers certain advantages in terms of the adjoint problem. In particular, within this approach the rate of change of the ice thickness occurs naturally as a model parameter, and hence singular behaviour of the sensitivity kernel for ice thickness is avoided. Within this work, by contrast, singularities of the ice kernel must be addressed directly. Nevertheless, the rate-formulation is not used within other GIA codes that account for laterally varying viscosity (e.g. Zhong *et al.* 2003; Latychev *et al.* 2005b; Steffen *et al.* 2006) and its adoption within them would require non-trivial modifications.

A further point is that the numerical implementation of the rate-formulation depends on explicit time stepping schemes that lack uniform stability. The largest time step is set by a constant fraction of the smallest relaxation time within the earth model. For calculations in realistic laterally heterogeneous models, this time step can be prohibitively small for simulations spanning the deglacial period. While there do exist stable time stepping schemes suitable for modelling quasi-static viscoelastic deformation (e.g. Simo & Hughes 1998; Bailey 2006), their implementation within the context of the rate-formulation is challenging and may not be possible. By moving towards a conventional formulation of the GIA forward problem, these numerical difficulties can be circumvented, and hence the application of adjoint methods within the GIA inverse problem made significantly more efficient.

2 SUMMARY OF THE FORWARD GIA PROBLEM

We begin by summarizing in suitable form the equations of motion for GIA. In doing this, we assume a Maxwell rheology and account for shoreline migration and rotational feedbacks. This discussion builds directly on Al-Attar & Tromp (2014), Crawford *et al.* (2018), and Al-Attar *et al.* (2024), with these works being in turn based on the earlier literature (e.g. Dahlen 1974; Peltier 1976; Farrell & Clark 1976; Mitrovica & Peltier 1991; Dahlen & Tromp 1998; Milne & Mitrovica 1998; Tromp & Mitrovica 1999; Mitrovica & Milne 2003; Kendall *et al.* 2005). For simplicity, we neglect the existence of a fluid outer core within the main text. The necessary details are provided within Appendix A where it is shown that the inclusion of fluid regions has no effect on the form of our main results.

2.1 Static loading on an elastic planet

We consider a non-rotating and isotropic elastic earth model that is initially in hydrostatic equilibrium. The earth model does not include an ocean or ice sheets, with these features later introduced through appropriate surface loads. Let M denote the volume of the earth model at equilibrium, and ∂M its surface which has outward unit normal $\hat{\mathbf{n}}$. The hydrostatic equilibrium condition requires that

$$\nabla p + \rho \nabla \Phi = 0, \tag{1}$$

where p, ρ, and Φ are, respectively, the pressure, density, and gravitational potential.

Suppose that the equilibrium state is disturbed by the application of a surface load, σ. The resulting deformation can be described by a displacement vector $\mathbf{u}$, along with an Eulerian perturbation, ϕ, to the gravitational potential. The linearized equations of motion can be written concisely in weak form as

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla \Phi + \phi')\, \sigma \, \mathrm{d}S = 0, \tag{2}$$

which is required to hold for arbitrary test functions $\mathbf{u}'$ and ϕ' (Al-Attar & Tromp 2014). Here, we have the bilinear form

$$\begin{aligned} \mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = {} & \int_M \kappa \, \nabla \cdot \mathbf{u} \, \nabla \cdot \mathbf{u}' \, \mathrm{d}^3\mathbf{x} + \int_M 2\mu \, \mathbf{d} : \mathbf{d}' \, \mathrm{d}^3\mathbf{x} + \frac{1}{2} \int_M \rho \left[\nabla(\mathbf{u} \cdot \nabla \Phi) \cdot \mathbf{u}' + \nabla(\mathbf{u}' \cdot \nabla \Phi) \cdot \mathbf{u} \right] \mathrm{d}^3\mathbf{x} \\ & - \frac{1}{2} \int_M \rho \left(\nabla \cdot \mathbf{u} \, \nabla \Phi \cdot \mathbf{u}' + \nabla \cdot \mathbf{u}' \, \nabla \Phi \cdot \mathbf{u} \right) \mathrm{d}^3\mathbf{x} + \int_M \rho \left(\nabla \phi \cdot \mathbf{u}' + \nabla \phi' \cdot \mathbf{u} \right) \mathrm{d}^3\mathbf{x} \\ & + \frac{1}{4\pi G} \int_{\mathbf{R}^3} \nabla \phi \cdot \nabla \phi' \, \mathrm{d}^3\mathbf{x}, \end{aligned} \tag{3}$$

associated with elastic and gravitational restoring forces. Within this expression, G is the gravitational constant, and κ and μ are the bulk and shear modulii. The linearized strain tensor, $\mathbf{e}$ is defined by

$$\mathbf{e} = \frac{1}{2} \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T \right], \tag{4}$$

while its deviatoric part is

$$\mathbf{d} = \mathbf{e} - \frac{1}{3} \mathrm{tr}(\mathbf{e}) \mathbf{1}, \tag{5}$$

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with tr(·) denoting the trace of a matrix and **1** the identity matrix. The term $\mathbf{d}'$ is defined in an identical manner with respect to the test function $\mathbf{u}'$. We note the symmetry

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = \mathcal{A}(\mathbf{u}', \phi' \,|\, \mathbf{u}, \phi), \tag{6}$$

that will be used repeatedly later on. As a final condition, we require that ϕ tends to zero at infinity, this serving to fix the arbitrary constant that can always be added to the potential.

2.2 Rigid body motions

A linearized rigid body motion takes the form

$$\mathbf{u} = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \tag{7}$$

with **a** and **b** constant vectors that describe, respectively, degrees of freedom associated with translation and rotation. The associated gravitational potential perturbation at a fixed spatial point is

$$\phi = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi. \tag{8}$$

For such fields it can be verified directly that

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = 0, \tag{9}$$

for any test functions $(\mathbf{u}', \phi')$. An immediate consequence is that the solution of the static elastic loading problem is defined only up to an arbitrary linearized rigid-body motion. This means that if $(\mathbf{u}, \phi)$ solves the loading problem for a given σ, then so do the fields

$$\mathbf{u} + \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi - (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi, \tag{10}$$

for any **a** and **b**. Assuming that the elastic earth model is gravitationally stable, it can be shown that the 6-D subspace of linearized rigid body motions comprises the only fields for which eq. (9) holds (e.g. Dahlen & Tromp 1998, sections 4.1.5 and 4.1.6).

Suppose that within eq. (2), we take as test functions

$$\mathbf{u}' = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi' = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi. \tag{11}$$

Using eqs (6) and (9), the first term within eq. (2) vanishes, and hence we are left with

$$\int_{\partial M} (\mathbf{u}' \cdot \nabla \Phi + \phi') \sigma \, \mathrm{d}S = 0, \tag{12}$$

as a necessary condition on the load in order for solutions of the problem to exist. In fact, it can readily be seen that $\mathbf{u}' \cdot \nabla \Phi + \phi' = 0$ for our chosen test functions, and hence the condition on the load is automatically met. It follows from the Fredholm alternative for elliptic partial differential equations that the above condition on the load is also sufficient for solutions of eq. (2) to exist (e.g. Marsden & Hughes 1983, chapter 6). We conclude that the static loading problem has solutions for any given load, with these solutions defined uniquely up to the addition of an arbitrary linearized rigid motion.

2.3 Rotational feedbacks

Surface loading of the Earth is associated with changes to its moment of inertia and hence to its angular velocity to conserve angular momentum. These rotational changes in turn generate centrifugal forces that modify the deformation, and thus feedbacks exist between load-induced deformation and rotational variations (e.g. Sabadini & Peltier 1981; Wu & Peltier 1984; Milne & Mitrovica 1998; Martinec & Hagedoorn 2005; Mitrovica *et al.* 2005; Mitrovica & Wahr 2011). Within GIA studies, it has been usual to account for rotational feedbacks in an approximate manner as summarized below. A notable feature of this approach is that it allows for calculations to be performed in spherically symmetric earth models which, until relatively recently, was all that was feasible.

Within the traditional approach to rotational feedbacks, static loading of a non-rotating model is still considered, but an additional force term is included to represent a centrifugal potential perturbation. Following the discussion in Al-Attar *et al.* (2024), the static elastic loading problem in eq. (2) is generalized to read

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla \Phi + \phi') \sigma \, \mathrm{d}S + \int_M \rho \, \mathbf{u}' \cdot \nabla \psi \, \mathrm{d}^3\mathbf{x} = 0, \tag{13}$$

with $(\mathbf{u}', \phi')$ again test functions, and where the perturbed centrifugal potential, ψ, is related to the perturbed angular velocity, $\boldsymbol{\omega}$, through

$$\psi = -(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega} \times \mathbf{x}), \tag{14}$$

with $\boldsymbol{\Omega}$ the equilibrium angular velocity. For given values of the load, σ, and centrifugal potential perturbation, ψ, the linearized equations of motion can be solved for the displacement vector **u**, and the gravitational potential perturbation, ϕ. Using these results along with the direct

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contribution of the surface load, the perturbation to the model's inertia tensor can be determined. From the inertia tensor perturbation the resultant change in the angular velocity can be found via conservation of angular momentum. The latter calculation depends on knowledge of the equilibrium inertia tensor, though we note that this tensor cannot, in general, be taken directly from the earth model in which the elastic calculations have been performed (e.g. Mitrovica *et al.* 2005). As shown in Al-Attar *et al.* (2024), the resulting relationship between the displacement, load, and perturbed angular velocity can be concisely written

$$\boldsymbol{\omega}' \cdot \mathbf{D} \cdot \boldsymbol{\omega} + \int_M \rho\, \mathbf{u} \cdot \nabla \psi'\, \mathrm{d}^3\mathbf{x} + \int_{\partial M} \sigma\, \psi'\, \mathrm{d}S = 0, \tag{15}$$

where $\boldsymbol{\omega}'$ is an arbitrary vector, $\psi' = -(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega}' \times \mathbf{x})$, and $\mathbf{D}$ the symmetric and invertible matrix

$$\mathbf{D} = \begin{pmatrix} C_3 - C_1 & 0 & 0 \\ 0 & C_3 - C_2 & 0 \\ 0 & 0 & -C_3 \end{pmatrix}, \tag{16}$$

with $C_1 \leq C_2 < C_3$ the principal moments of inertia assumed. Eqs (13) and (15) can be combined into a single weak form

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla \Phi + \phi' + \psi')\, \sigma\, \mathrm{d}S + \int_M \rho \left(\mathbf{u}' \cdot \nabla \psi + \mathbf{u} \cdot \nabla \psi'\right) \mathrm{d}^3\mathbf{x} + \boldsymbol{\omega}' \cdot \mathbf{D} \cdot \boldsymbol{\omega} = 0, \tag{17}$$

which is to hold for arbitrary test functions $\mathbf{u}'$, ϕ', and $\boldsymbol{\omega}'$. While these equations could be solved for $\mathbf{u}$, ϕ, and $\boldsymbol{\omega}$ simultaneously, it is typical for an iterative approach to be applied with an initial guess for $\boldsymbol{\omega}$ successively refined.

2.4 Rigid body motions (again)

We need to reassess the issue of uniqueness and existence within the loading problem now that rotational feedbacks have been included. As with the earlier static loading problem, for given (σ, ψ) we see from eq. (13) that solutions can only be defined up to a linearized rigid body motion. If we take

$$\mathbf{u}' = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi' = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi, \tag{18}$$

within eq. (13) we arrive at the equality

$$\int_M (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \rho\, \nabla \psi\, \mathrm{d}^3\mathbf{x} = 0, \tag{19}$$

as a necessary and sufficient condition for solutions to exist. Assuming for the moment that this condition is met, we can find a solution $(\mathbf{u}, \phi)$ of the static elastic equations determined up to a linearized rigid body motion. The matrix $\mathbf{D}$ in eq. (15) is invertible, and hence this equation has a unique solution, $\boldsymbol{\omega}$, for given $(\mathbf{u}, \sigma)$. Because, however, $\mathbf{u}$ is only determined by eq. (13) up to a linearized rigid body motion, we need to ask how this non-uniqueness manifests in $\boldsymbol{\omega}$. If, within, eq. (15) we replace the displacement vector by $\mathbf{u} + \mathbf{a} + \mathbf{b} \times \mathbf{x}$, it can be seen that the solution $\boldsymbol{\omega}$ is unchanged if and only if

$$\int_M (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \rho\, \nabla \psi'\, \mathrm{d}^3\mathbf{x} = 0, \tag{20}$$

where we recall that ψ' is the centrifugal potential perturbation associated with an arbitrary angular velocity $\boldsymbol{\omega}'$; clearly this requirement is equivalent to the existence condition in eq. (19). Summarizing the above discussion, we have shown that solutions of the elastostatic loading problem with rotational feedbacks exist so long as eq. (20) holds for arbitrary $\boldsymbol{\omega}'$. When this condition is met, the solutions are defined uniquely modulo the transformation

$$\mathbf{u} \mapsto \mathbf{u} + \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi \mapsto \phi - (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi, \quad \boldsymbol{\omega} \mapsto \boldsymbol{\omega}, \tag{21}$$

for arbitrary constant vectors $\mathbf{a}$ and $\mathbf{b}$.

It remains to determine whether eq. (20) holds in practice. To do this, consider eq. (15) in the case that $\sigma = 0$. We then have

$$\boldsymbol{\omega}' \cdot \mathbf{D} \cdot \boldsymbol{\omega} + \int_M \rho\, \mathbf{u} \cdot \nabla \psi'\, \mathrm{d}^3\mathbf{x} = 0, \tag{22}$$

which gives the perturbed angular velocity, $\boldsymbol{\omega}$, associated with the displacement vector, $\mathbf{u}$. If we take $\mathbf{u} = \mathbf{a} + \mathbf{b} \times \mathbf{x}$ within this equation and assume that eq. (20) holds, then we see that $\boldsymbol{\omega} = \mathbf{0}$. It follows that eq. (20) is equivalent to the invariance of the inertia tensor for the equilibrium earth model under arbitrary linearized rigid body motions, and hence this tensor must be isotropic. Note that this is a statement about the earth model assumed within the elastostatic calculations, and not about the equilibrium inertia tensor used when calculating perturbations to the angular velocity via conservation of angular momentum.

We have shown that the standard theory of rotational feedbacks implicitly assumes that model in which the elastic calculations are performed has an isotropic moment of inertia tensor. To our knowledge, this is an original observation. This rotational theory was developed within the context of spherically symmetric earth models for which the requirement is met, but it remains in use within more recent calculations in laterally heterogeneous earth models and here problems could emerge. In practice, however, such calculations are focused on the effect

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of large lateral viscosity variations within the mantle, with smaller lateral variations in the model's shape or density typically being ignored. More sophisticated theories for rotational feedbacks have been developed based upon perturbations to a rotating equilibrium state, and there then exist no implicit constraints on the structure of the earth model. Work in this direction includes Martinec & Hagedoorn (2005, 2014), along with the recent study by Maitra & Al-Attar (2024) which includes a complete non-linear theory for quasi-static loading within a variably rotating earth model. Within this paper, we nonetheless retain the traditional rotational theory because it is the most widely used and because it remains appropriate in most cases of practical interest.

2.5 Viscoelastic relaxation

The equations stated so far apply to the static loading of an elastic earth model, but they can be readily extended to the case of quasi-static viscoelasticity. Within GIA studies, a Maxwell solid rheology is most commonly assumed, and we focus on this case within this paper. The incorporation of more complex linear or non-linear rheolgies within both the forward and adjoint theory is not difficult; see Crawford *et al.* (2017) for results along these lines in the context of post-seismic deformation. As discussed by Al-Attar & Tromp (2014), the linearized stress tensor for a Maxwell solid is

$$\mathbf{T} = \kappa \, \nabla \cdot \mathbf{u} \, \mathbf{1} + 2\mu \, (\mathbf{d} - \mathbf{m}), \tag{23}$$

where $\mathbf{m}$ is an internal variable that satisfies the following differential equation

$$\dot{\mathbf{m}} + \frac{1}{\tau}(\mathbf{m} - \mathbf{d}) = \mathbf{0}, \tag{24}$$

with τ the relaxation time. Such an internal variable approach to viscoelasticity is widely used within the engineering literature (e.g. Simo & Hughes 1998; Holzapfel 2000) and it is equivalent to other time-domain schemes that have been applied within the context of GIA (e.g. Hanyk *et al.* 1995; Zhong *et al.* 2003; Latychev *et al.* 2005b; Bailey 2006). Generalizing eq. (17) to account for Maxwell viscoelasticity, we arrive at the appropriate weak form

$$\begin{aligned} &\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') - \int_M 2\mu \, \mathbf{m} : \mathbf{d}' \, \mathrm{d}^3\mathbf{x} + \int_{\partial M} (\mathbf{u}' \cdot \nabla \Phi + \phi' + \psi') \, \sigma \, \mathrm{d}S \\ &+ \int_M \rho \left(\mathbf{u}' \cdot \nabla \psi + \mathbf{u} \cdot \nabla \psi' \right) \mathrm{d}^3\mathbf{x} + \boldsymbol{\omega}' \cdot \mathbf{D} \cdot \boldsymbol{\omega} + \int_M 2\mu \, (\tau \, \dot{\mathbf{m}} + \mathbf{m} - \mathbf{d}) : \mathbf{m}' \, \mathrm{d}^3\mathbf{x} = 0, \end{aligned} \tag{25}$$

where an additional test function, $\mathbf{m}'$, has been introduced to enforce the evolution equation for the internal variable. Note that this equality is required to hold for all times, but with test functions being time-independent.

Within this problem it is necessary, at each time step, to solve an elastostatic equation forced by the load, the centrifugal potential perturbation, and a body force associated with the internal variable. Physically, the latter force represents the viscoelastic relaxation of stress. Having done this, we can compute $\mathbf{d}$ and then use the evolution equation for $\mathbf{m}$ to advance the system in time. This is all done subject to the initial condition

$$\mathbf{m}(\mathbf{x}, t_0) = \mathbf{0}. \tag{26}$$

Because the internal variable couples to the deformation only through the deviatoric strain, our previous discussion of uniqueness and existence carries over immediately, with solutions of the problem defined up to the following transformations

$$\mathbf{u} \mapsto \mathbf{u} + \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi \mapsto \phi - (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla \Phi, \quad \boldsymbol{\omega} \mapsto \boldsymbol{\omega}, \quad \mathbf{m} \mapsto \mathbf{m}, \tag{27}$$

for any spatially constant vectors $\mathbf{a}$ and $\mathbf{b}$.

Other approaches to viscoelasticity in the context of GIA are possible. Boltzmann's superposition principle provides a general convolutional form for the constitutive relation within linearized viscoelasticity (e.g. Coleman & Noll 1961), and from this one can also arrive at the so-called correspondence principle using Laplace-transform methods (e.g. Peltier 1974). Within the application of adjoint methods to GIA, our view is that time-domain formulations for viscoelasticity are most appropriate. This is because of their near ubiquitous use within codes for modelling GIA in laterally heterogeneous earth models. Note, however, that in the adjoint theory of Martinec *et al.* (2015), the general convolutional form for linearized viscoelasticity was used.

2.6 Gravitationally self-consistent water loads

To complete specification of the GIA forward problem, we need to link the deformation of the earth model to changes in sea level. This leads us to the sea level theory of Farrell & Clark (1976) that has been subsequently modified to account for shoreline migration and rotational feedbacks (e.g. Milne & Mitrovica 1998; Mitrovica & Milne 2003; Kendall *et al.* 2005). The present discussion follows closely that in Crawford *et al.* (2018) and Al-Attar *et al.* (2024).

The total surface load at given time is the sum of the ocean and ice loads and can be written

$$\rho_w \, C \, SL + \rho_i \, (1 - C) \, I, \tag{28}$$

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where ρ_w is the water density, C the ocean function, SL the sea level, ρ_i the ice density, and I the ice thickness. Underlying this expression are a host of standard definitions and assumptions that we briefly summarize. First, we are using a hydrostatic theory for the oceans, and hence define the sea level to be the signed distance from the solid surface to the equipotential of gravity on which the sea surface lies. To allow for ice shelves we apply a simple hydrostatic balance, with this leading to the following definition of the ocean function

$$C(\mathbf{x}, t) = \begin{cases} 1 & \rho_w\, SL(\mathbf{x}, t) - \rho_i\, I(\mathbf{x}, t) > 0 \\ 0 & \rho_w\, SL(\mathbf{x}, t) - \rho_i\, I(\mathbf{x}, t) \leq 0 \end{cases}. \tag{29}$$

This function equals one in the oceans and zero otherwise. For later convenience, we write

$$\mathcal{O}_t = \{\mathbf{x} \in \partial M \,|\, \rho_w\, SL(\mathbf{x}, t) - \rho_i\, I(\mathbf{x}, t) > 0\}, \tag{30}$$

for the subset of ∂M covered by oceans at time t, and

$$\partial\mathcal{O}_t = \{\mathbf{x} \in \partial M \,|\, \rho_w\, SL(\mathbf{x}, t) - \rho_i\, I(\mathbf{x}, t) = 0\}, \tag{31}$$

for the associated shoreline (including the grounding line in regions with ice shelves).

Within the linearized equations of motion, it is not the total load that is relevant but the change in the load since an initial time, t_0, when the system is assumed be in equilibrium. We can, therefore, write the load occurring within eq. (17) as

$$\sigma = \rho_w\, C\, SL + \rho_i\, (1 - C)\, I - \rho_w\, C_0\, SL_0 - \rho_i\, (1 - C_0)\, I_0, \tag{32}$$

where the subscript zero is used to denote values at the initial time. In terms of this load, conservation of mass between the oceans and ice sheets can be conveniently expressed as

$$\int_{\partial M} \sigma \, \mathrm{d}S = 0. \tag{33}$$

Finally, the sea level is related to the solid earth deformation through

$$SL = SL_0 - \frac{1}{g}(\mathbf{u} \cdot \nabla\Phi + \phi + \psi) + \frac{\Phi_g}{g}, \tag{34}$$

where g is the acceleration due to gravity at the surface and Φ_g is a spatially constant field whose value is fixed through eq. (33). Note that the dependence of sea level on the solid earth deformation is invariant under linearized rigid body motions. Thus, the sea level is uniquely determined from solution of the equations of motion. In what follows, it will sometimes be useful to write

$$SL_1 = SL - SL_0, \tag{35}$$

for the change in sea level since the initial time.

Within eq. (32), the load, σ, depends on the sea level, which is one of the variables we wish to determine. The sea level is, in turn, related to the load-induced deformation through eq. (34). Thus, the forces within the problem depend on its solution, and hence we arrive at the sea level equation of Farrell & Clark (1976). In practice, this problem can be solved efficiently using a simple iterative scheme. The assumed sea level at each stage is used to determine a load, from this load the resulting deformation, and hence an improved estimate for the sea level. Convergence of this scheme is typically obtained in only a handful of iterations.

2.7 Summary of the forward problem

For convenience, we gather together the complete set of equations for the GIA forward problem within a unified weak form:

$$\begin{aligned} &\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') - \int_M 2\mu\mathbf{m} : \mathbf{d}'\, \mathrm{d}^3\mathbf{x} + \int_{\partial M} \left[\mathbf{u}' \cdot \nabla\Phi + \phi' + \psi'\right] \sigma \, \mathrm{d}S \\ &+ \int_M \rho \left[\mathbf{u}' \cdot \nabla\psi + \mathbf{u} \cdot \nabla\psi'\right] \mathrm{d}^3\mathbf{x} + \boldsymbol{\omega}' \cdot \mathbf{D} \cdot \boldsymbol{\omega} + \int_M 2\mu\, (\tau\, \dot{\mathbf{m}} + \mathbf{m} - \mathbf{d}) : \mathbf{m}'\, \mathrm{d}^3\mathbf{x} \\ &+ \int_{\partial M} [\sigma - \rho_w\, C\, SL - \rho_i\, (1 - C)\, I + \rho_w\, C_0\, SL_0 + \rho_i\, (1 - C_0)\, I_0]\, g\, SL_1' \, \mathrm{d}S \\ &+ \int_{\partial M} \left(g\, SL_1 + \mathbf{u} \cdot \nabla\Phi + \phi + \psi - \Phi_g\right) \sigma' \, \mathrm{d}S - \Phi_g' \int_{\partial M} \sigma \, \mathrm{d}S = 0. \end{aligned} \tag{36}$$

Here new test functions, SL_1', σ', and Φ_g' associated, respectively, with eqs (32), (34), and (33) have been introduced, while we recall that the stated equality is to hold for all possible test-functions. Note that in combining the various equations certain sign choices have been made and scalar factors included so as to maximize the symmetry of the resulting expression.

Writing the forward problem in weak form is convenient both notationally and practically. It is this form of the equations that is necessary within the application of finite element methods to GIA, while the use of weak forms simplifies the derivation and statement of the adjoint equations within the next section. If required, the corresponding strong form of the equations of motion can be readily obtained using integration by parts. Doing this would yield the usual equations for quasi-static momentum balance, Newtonian gravitation, and all associated boundary conditions (e.g. Dahlen 1974; Tromp & Mitrovica 1999).

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3 SENSITIVITY KERNELS AND THE FIRST-ORDER ADJOINT GIA PROBLEM

Within this section, we apply the first-order adjoint method to obtain sensitivity kernels for the GIA problem with respect to mantle viscosity, ice thickness, and initial sea level (or, equivalent, palaeo-topography). Later, when we discuss the second-order adjoint method, it will be necessary to distinguish between first- and second-order adjoint variables and equations. For the moment, however, it is understood that in saying, for example, the "adjoint equations" we mean "first-order adjoint equations".

3.1 Objective functionals and sensitivity kernels

Consider a real-valued *objective functional*

$$J = J(SL, \mathbf{u}, \phi, \boldsymbol{\omega}), \tag{37}$$

defined in terms of the sea level, SL, surface values of $\mathbf{u}$ and ϕ, and the perturbed angular velocity, $\boldsymbol{\omega}$, over a time interval $\mathcal{I} = [t_0, t_1]$. Through a GIA simulation, the arguments, $(SL, \mathbf{u}, \phi, \boldsymbol{\omega})$, of J can be calculated given suitable model parameters, and hence the value of this functional determined. For definiteness, we will focus on the model parameters of the widest interest within the GIA community:

(i) viscosity, $\eta = \mu\tau$, defined in M;
(ii) ice sheet thickness, I, defined on $\partial M \times \mathcal{I}$;
(iii) initial sea level, SL_0, at time t_0, defined on ∂M.

Other properties of the Earth such as elastic modulii, density, or topography on internal boundaries might also be considered. Such quantities are, however, best constrained through seismological studies and are likely to have comparatively little impact in the context of GIA (e.g. Austermann *et al.* 2021). We can now define a *reduced objective functional* through

$$\hat{J}(\eta, I, SL_0) = J(SL, \mathbf{u}, \phi, \boldsymbol{\omega}), \tag{38}$$

where it is understood that the arguments on the right-hand side have been determined from those on the left through the solution of the GIA forward problem.

If the model parameters are perturbed, the first-variation of $\hat{J}$ can be written as

$$\delta\hat{J} = \int_M D_\eta \hat{J}\, \delta\eta \, \mathrm{d}^3\mathbf{x} + \int_{\mathcal{I}} \int_{\partial M} D_I \hat{J}\, \delta I \, \mathrm{d}S \, \mathrm{d}t + \int_{\partial M} D_{SL_0} \hat{J}\, \delta SL_0 \, \mathrm{d}S, \tag{39}$$

with the functional derivatives $D_\eta \hat{J}$, $D_I \hat{J}$, and $D_{SL_0} \hat{J}$ defined through this expression. These functional derivatives are equivalently called *sensitivity kernels*, and were denoted by K_η, K_I, and K_{SL_0} within Crawford *et al.* (2018).

Recall that solutions of the GIA forward problem are only unique up to a time-dependent linearized rigid body motion. In order for the objective functional to be well-defined we require that J be invariant with respect to such transformations, this meaning

$$J(SL, \mathbf{u} + \mathbf{a} + \mathbf{b} \times \mathbf{x}, \phi - (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla\Phi, \boldsymbol{\omega}) = J(SL, \mathbf{u}, \phi, \boldsymbol{\omega}), \tag{40}$$

for all spatially constant vectors $\mathbf{a}$ and $\mathbf{b}$. As an example, any J defined solely in terms of the sea level satisfies this condition. This is because sea level is the *relative* distance between two surfaces. Conversely, the value of the radial displacement at a given point and time provides an example of an invalid objective functional.

3.2 PDE constrained optimization via the Lagrangian method

The first-variation of δJ with respect to $(SL, \mathbf{u}, \phi, \boldsymbol{\omega})$ can be written

$$\delta J = \int_{\mathcal{I}} \left\{ \int_{\partial M} \left(D_{SL} J\, \delta SL + D_{\mathbf{u}} J \cdot \delta\mathbf{u} + D_\phi J\, \delta\phi \right) \mathrm{d}S + D_{\boldsymbol{\omega}} J \cdot \delta\boldsymbol{\omega} \right\} \mathrm{d}t, \tag{41}$$

for appropriate functional derivatives. The invariance condition for the objective functional in eq. (40) implies that

$$\int_{\partial M} \left(D_{\mathbf{u}} J - D_\phi J\, \nabla\Phi \right) \cdot (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \, \mathrm{d}S = 0, \tag{42}$$

with $\mathbf{a}$ and $\mathbf{b}$ arbitrary constant vectors.

To determine the variation of $\hat{J}$ with respect to the model parameters, (η, I, SL_0), we apply the Lagrangian method for PDE (partial differential equation) constrained optimization (e.g. Tröltzsch 2005, section 2.10). At the outset, it will be useful to introduce a *time-reversal operator*, $\mathcal{T}$. Let f be a function defined on the time interval $\mathcal{I} = [t_0, t_1]$, and set

$$(\mathcal{T} f)(t) = f(t_1 - t + t_0). \tag{43}$$

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This definition applies identically to scalar-, vector-, and tensor-valued functions and to those that depend on position also. It is clear that $\mathcal{T}$ is involutary (i.e., $\mathcal{T}^{-1} = \mathcal{T}$), that it anticommutes with time-differentiation, meaning

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{T}f) = -\mathcal{T}\left(\frac{\mathrm{d}f}{\mathrm{d}t}\right), \tag{44}$$

and that it is self-adjoint relative to the standard inner product on $L^2(\mathcal{I})$. In the case of scalar-valued functions, the latter condition implies

$$\int_{\mathcal{I}} (\mathcal{T}f)\, g \,\mathrm{d}t = \int_{\mathcal{I}} f\, (\mathcal{T}g)\,\mathrm{d}t, \tag{45}$$

for arbitrary f and g, while the extension to vector- and tensor-valued functions is obvious. We now define a Lagrangian for the problem

$$\begin{aligned} L = J + \int_{\mathcal{I}} \Big\{ & \mathcal{A}\left(\mathbf{u}, \phi \,|\, \mathcal{T}\mathbf{u}^\dagger, \mathcal{T}\phi^\dagger\right) - \int_M 2\mu \mathbf{m} : \left(\mathcal{T}\mathbf{d}^\dagger\right) \mathrm{d}^3\mathbf{x} + \int_{\partial M} \left[\left(\mathcal{T}\mathbf{u}^\dagger\right) \cdot \nabla\Phi + \mathcal{T}\phi^\dagger + \mathcal{T}\psi^\dagger\right] \sigma \,\mathrm{d}S \\ & + \int_M \rho \left[\left(\mathcal{T}\mathbf{u}^\dagger\right) \cdot \nabla\psi + \mathbf{u} \cdot \nabla\left(\mathcal{T}\psi^\dagger\right)\right] \mathrm{d}^3\mathbf{x} + \left(\mathcal{T}\boldsymbol{\omega}^\dagger\right) \cdot \mathbf{D} \cdot \boldsymbol{\omega} + \int_M 2\mu\, \left(\tau\, \dot{\mathbf{m}} + \mathbf{m} - \mathbf{d}\right) : \left(\mathcal{T}\mathbf{m}^\dagger\right) \mathrm{d}^3\mathbf{x} \\ & + \int_{\partial M} \left[\sigma - \rho_w\, C\, SL - \rho_i\, (1 - C)\, I + \rho_w\, C_0\, SL_0 + \rho_i\, (1 - C_0)\, I_0\right] g \left(\mathcal{T}SL_1^\dagger\right) \mathrm{d}S \\ & + \int_{\partial M} \left(g\, SL_1 + \mathbf{u} \cdot \nabla\Phi + \phi + \psi - \Phi_g\right) \left(\mathcal{T}\sigma^\dagger\right) \mathrm{d}S - \left(\mathcal{T}\Phi_g^\dagger\right) \int_{\partial M} \sigma \,\mathrm{d}S \Big\} \,\mathrm{d}t. \end{aligned} \tag{46}$$

Here, J is the objective functional of interest, while the remaining terms are associated with the constraint that the *state variables*, ($\mathbf{u}$, ϕ, $\boldsymbol{\omega}$, $\mathbf{m}$, SL, σ, Φ_g), satisfy the forward problem. For each state variable there is a corresponding *adjoint state variable* indicated by the same symbol but with the addition of a dagger superscript. It is worth emphasizing that within this definition the state and adjoint state variables are independent of one another, with the dagger notation simply providing a labelling scheme for the new variables. Time reversal of the adjoint state variables within the Lagrangian is not necessary but has been done for later convenience.

The Lagrange multiplier theorem shows that if L is stationary with respect to both the state variables and the adjoint state variables then functional derivatives of L and $\hat{J}$ with respect to the model parameters are equal. By construction, L being stationary with respect to the adjoint state variables implies that the state variables solve the forward problem. Indeed, taking the necessary variation and requiring the equality hold for each time separately, we arrive at the unified weak form of the GIA problem stated within eq. (36).

Requiring that L is stationary with respect to the state variables, on the other hand, defines a new system of equations that must be satisfied by the adjoint state variables. To obtain these latter equations, we first set the variation of L with respect to ($\mathbf{u}$, ϕ, $\boldsymbol{\omega}$, $\mathbf{m}$) equal to zero. Using eq. (41), this implies

$$\begin{aligned} & \int_{\mathcal{I}} \Big\{ \int_{\partial M} \left(D_{\mathbf{u}} J \cdot \delta\mathbf{u} + D_\phi J\, \delta\phi\right) \mathrm{d}S + D_{\boldsymbol{\omega}} J \cdot \delta\boldsymbol{\omega} + \mathcal{A}(\delta\mathbf{u}, \delta\phi \,|\, \mathcal{T}\mathbf{u}^\dagger, \mathcal{T}\phi^\dagger) - \int_M 2\mu\delta\mathbf{m} : (\mathcal{T}\mathbf{d}^\dagger)\, \mathrm{d}^3\mathbf{x} \\ & + \int_M \rho \left[\left(\mathcal{T}\mathbf{u}^\dagger\right) \cdot \nabla\delta\psi + \delta\mathbf{u} \cdot \nabla(\mathcal{T}\psi^\dagger)\right] \mathrm{d}^3\mathbf{x} + (\mathcal{T}\boldsymbol{\omega}^\dagger) \cdot \mathbf{D} \cdot \delta\boldsymbol{\omega} + \int_M 2\mu\, (\tau\, \delta\dot{\mathbf{m}} + \delta\mathbf{m} - \delta\mathbf{d}) : (\mathcal{T}\mathbf{m}^\dagger)\, \mathrm{d}^3\mathbf{x} \\ & + \int_{\partial M} (\delta\mathbf{u} \cdot \nabla\Phi + \delta\phi + \delta\psi)(\mathcal{T}\sigma^\dagger)\, \mathrm{d}S \Big\} \,\mathrm{d}t = 0. \end{aligned} \tag{47}$$

Integrating by parts with respect to time and using the properties of $\mathcal{A}$, $\mathbf{D}$, and $\mathcal{T}$, we find that the above equality reduces to

$$\begin{aligned} & \mathcal{A}(\mathbf{u}^\dagger, \phi^\dagger \,|\, \delta\mathbf{u}, \delta\phi) - \int_M 2\mu\, \mathbf{m}^\dagger : \delta\mathbf{d}\, \mathrm{d}^3\mathbf{x} + \int_{\partial M} (\delta\mathbf{u} \cdot \nabla\Phi + \delta\phi + \delta\psi)\, \sigma^\dagger \,\mathrm{d}S + \int_{\partial M} \left[(\mathcal{T}D_{\mathbf{u}} J) \cdot \delta\mathbf{u} + \left(\mathcal{T}D_\phi J\right) \delta\phi\right] \mathrm{d}S \\ & + \int_M \rho\, (\mathbf{u}^\dagger \cdot \nabla\delta\psi + \delta\mathbf{u} \cdot \nabla\psi^\dagger)\, \mathrm{d}^3\mathbf{x} + \delta\boldsymbol{\omega} \cdot \left[\mathbf{D} \cdot \boldsymbol{\omega}^\dagger + (\mathcal{T}D_{\boldsymbol{\omega}} J)\right] + \int_M 2\mu\, \left(\tau\dot{\mathbf{m}}^\dagger + \mathbf{m}^\dagger - \mathbf{d}^\dagger\right) : \delta\mathbf{m}\, \mathrm{d}^3\mathbf{x} = 0, \end{aligned} \tag{48}$$

where ($\delta\mathbf{u}$, $\delta\phi$, $\delta\boldsymbol{\omega}$, $\delta\mathbf{m}$) are now arbitrary time-independent test functions, along with the initial condition

$$\mathbf{m}^\dagger(\mathbf{x}, t_0) = \mathbf{0}. \tag{49}$$

Variation of L with respect to σ and Φ_g are trivial, yielding the following relations

$$SL_1^\dagger = -\frac{1}{g}\left(\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger + \psi^\dagger\right) + \frac{\Phi_g^\dagger}{g}, \quad \int_{\partial M} \sigma^\dagger \,\mathrm{d}S = 0. \tag{50}$$

In varying the Lagrangian with respect to sea level, we first note that SL_0 is a parameter in the problem, and so is fixed within this variation. This means that it is not SL but the sea level change, $SL_1 = SL - SL_0$, that undergoes arbitrary variation (it is for this reason we called the corresponding adjoint variable $SL_1^\dagger$). The other complication is that the ocean function depends on the sea level, and hence must be varied as part of the calculation. To do this, we recall the following expression

$$\delta C = \frac{\rho_w \delta SL_1}{\rho_w \partial_\perp SL - \rho_i \partial_\perp I} \delta_{\partial\mathcal{O}_t}, \tag{51}$$

for the first variation of the ocean function with respect to a change in sea level (Crawford *et al.* 2018, eq. A30). Here $\partial_\perp$ denotes the directional derivative perpendicular to the shoreline oriented towards the oceans, and $\delta_{\partial\mathcal{O}_t}$ is the Dirac line distribution on the shoreline. The action of

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this distribution on a smooth test function, f, is given by

$$\int_{\partial M} f\,\delta_{\partial\mathcal{O}_t}\,\mathrm{d}S = \int_{\partial\mathcal{O}_t} f\,\mathrm{d}l, \tag{52}$$

with $\mathrm{d}l$ the standard line element. A key simplification within the derivation is that the singular term δC is multiplied by the smooth function $\rho_w SL - \rho_i I$ which, by definition, vanishes on $\partial\mathcal{O}_t$. The remainder of the calculation is easy, with the result being

$$\sigma^\dagger = \rho_w\, C^\dagger\, SL_1^\dagger - \frac{1}{g}\mathcal{T} D_{SL} J, \tag{53}$$

where we have set $C^\dagger = \mathcal{T}C$ for convenience.

The existence and uniqueness for solutions for the adjoint problem can be readily analysed. First, it is clear that solutions, $(\mathbf{u}^\dagger, \phi^\dagger)$ of the elastostatic subproblem are defined only up to a linearized rigid body motion, but that this non-uniqueness does not propagate through to the other adjoint variables. The existence condition for the adjoint problem can be obtained by taking the test functions, $(\delta\mathbf{u}, \delta\phi)$, to be a linearized rigid body motion, this leading to the equality

$$\int_{\partial M} \left(D_{\mathbf{u}} J - D_\phi J\,\nabla\Phi\right)\cdot(\mathbf{a} + \mathbf{b}\times\mathbf{x})\,\mathrm{d}S + \int_M \rho\,(\mathbf{a}+\mathbf{b}\times\mathbf{x})\cdot\nabla\psi^\dagger\,\mathrm{d}^3\mathbf{x} = 0, \tag{54}$$

for arbitrary constant vectors $\mathbf{a}$ and $\mathbf{b}$. The first integral vanishes because of the assumed invariance of J, while we have seen that the second integral must be zero in order for the forward problem to have solutions. Thus, so long as the objective functional is physically well defined the adjoint problem admits solutions.

3.3 Practical implementation

The adjoint GIA problem is nearly identical in form to the forward problem. This is useful because it means that a code that can solve the forward problem can also solve the adjoint problem with minimal modifications. There are just two areas of difference. First, the adjoint problem involves more general force terms. In the forward problem, there is only the applied ice load. But in the adjoint problem there are four force terms, each being expressed in terms of a functional derivative of J with respect to one of its arguments. The term associated with $D_{SL}J$ plays a role similar to the ice load within the forward problem, while the term due to $D_{\boldsymbol{\omega}}J$ acts like an angular momentum perturbation when solving for $\boldsymbol{\omega}^\dagger$. Finally, the force terms associated with $D_{\mathbf{u}}J$ and $D_\phi J$ correspond to inhomogeneous boundary conditions on the surface of the earth model. For the convenience of the reader, we now summarize these boundary conditions explicitly. To put these results in context, we first recall that within the forward GIA problem the relevant boundary conditions for $\mathbf{u}$ and ϕ on ∂M are

$$\hat{\mathbf{n}}\cdot(\kappa\nabla\cdot\mathbf{u} + 2\mu\mathbf{d}) = -\sigma\nabla\Phi + \hat{\mathbf{n}}\cdot 2\mu\mathbf{m}, \tag{55}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi\right]_-^+ - \rho\hat{\mathbf{n}}\cdot\mathbf{u} = \sigma, \tag{56}$$

where $[\cdot]_-^+$ denotes a jump in a quantity across ∂M in the direction of the outward unit normal. Within the adjoint problem, these conditions are generalized to

$$\hat{\mathbf{n}}\cdot\left(\kappa\nabla\cdot\mathbf{u}^\dagger + 2\mu\mathbf{d}^\dagger\right) = -\sigma^\dagger\nabla\Phi + \hat{\mathbf{n}}\cdot 2\mu\mathbf{m}^\dagger - \mathcal{T}D_{\mathbf{u}}J, \tag{57}$$

$$\left[(4\pi G)^{-1}\hat{\mathbf{n}}\cdot\nabla\phi^\dagger\right]_-^+ - \rho\hat{\mathbf{n}}\cdot\mathbf{u}^\dagger = \sigma^\dagger + \mathcal{T}D_\phi J. \tag{58}$$

These two additional terms within the surface boundary conditions are the *only* differences between the forward and adjoint elastostatic subproblems, and hence a code that can solve the forward problem can be readily adapted to solve the adjoint one.

The remaining difference between the forward and adjoint GIA problems is that the ocean function, $C^\dagger$, occurring within the adjoint problem is not determined dynamically. Instead, it is the time-reversed ocean function obtained from solution of the forward problem. This means that the adjoint problem is linear but with time-dependent coefficients.

If shoreline migration is neglected within the forward problem, then the load reduces to

$$\sigma = \rho_w\, C\, SL_1 + \rho_i\,(1-C)\,I_1, \tag{59}$$

where SL_1 and I_1 denote, respectively, the change in sea level and ice thickness since t_0. The forward problem is then linear, and the adjoint equations differ from the forward problem only through the inclusion of more general force terms.

3.4 Expressions for the sensitivity kernels

We can now derive expressions for the sensitivity kernels, $D_\eta\hat{J}$, $D_I\hat{J}$, and $D_{SL_0}\hat{J}$ by varying the Lagrangian with respect to the model parameters. The first variation of $\hat{J}$ with respect to η is given by

$$\delta\hat{J} = \int_{\mathcal{I}}\int_M 2\delta\eta\,\dot{\mathbf{m}} : (\mathcal{T}\mathbf{m}^\dagger)\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t, \tag{60}$$

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where we have used $\eta = \mu\,\tau$ and recall that the shear modulus, μ, is fixed. Using the evolution equation for $\mathbf{m}$, it follows that the viscosity kernel can be written as

$$D_\eta \hat{J} = \int_{\mathcal{I}} \frac{2\mu}{\eta}\,(\mathbf{d} - \mathbf{m}) : (\mathcal{T}\mathbf{m}^\dagger)\,\mathrm{d}t. \tag{61}$$

Turning to the kernel for ice thickness, we need to account for changes in the ocean function when taking variations of the Lagrangian. To do this, we again apply eq. (A30) of Crawford *et al.* (2018) which, in this instance, takes the form

$$\delta C = \frac{-\rho_w \delta I}{\rho_w \partial_\perp SL - \rho_i \partial_\perp I}\delta_{\partial\mathcal{O}_t}. \tag{62}$$

As previously, singular terms on the shoreline cancel, and we are left with

$$\delta \hat{J} = -\int_{\mathcal{I}}\int_{\partial M} \rho_i\, g\,[(1-C)\delta I - (1-C_0)\delta I_0]\left(\mathcal{T}SL_1^\dagger\right)\,\mathrm{d}S\,\mathrm{d}t. \tag{63}$$

Using a Dirac delta function, we can identify the ice kernel as

$$D_I \hat{J} = -\rho_i\, g\left[(1-C)\,\left(\mathcal{T}SL_1^\dagger\right) - (1-C_0)\int_{\mathcal{I}} \mathcal{T}SL_1^\dagger\,\mathrm{d}t\;\delta(t-t_0)\right]. \tag{64}$$

The derivation of the kernel for initial sea level proceeds similarly, leading to the expression

$$D_{SL_0}\hat{J} = \int_{\mathcal{I}}\left[D_{SL}J - \rho_w g\,(C-C_0)\left(\mathcal{T}SL_1^\dagger\right)\right]\mathrm{d}t. \tag{65}$$

The sensitivity kernels are expressed as combinations of the forward and adjoint state variables. We know the displacement vector and gravitational potential perturbation within the forward and adjoint problems are only defined up to a linearized rigid body motion. From the above expressions it can, however, be readily seen that the kernels are invariant under such transformations, and hence uniquely defined.

3.5 An alternate form for the viscosity kernel

Within this paper, we have made use of an internal variable method to account for the Maxwell rheology. While this approach is widely used and easy to implement (e.g. Simo & Hughes 1998; Holzapfel 2000), a range of equivalent methods have been considered within the GIA literature. For the convenience of a reader hoping to implement the adjoint method within their own codes, we now present an alternative expression for the viscosity kernel which is independent of the internal variable formalism.

First, we recall that the stress, $\mathbf{T}$, takes the form

$$\mathbf{T} = \kappa\,\nabla\cdot\mathbf{u}\,\mathbf{1} + 2\mu\,(\mathbf{d}-\mathbf{m}), \tag{66}$$

and hence the deviatoric component of the stress, $\boldsymbol{\tau}$, can be expressed as

$$\boldsymbol{\tau} = 2\mu\,(\mathbf{d}-\mathbf{m}). \tag{67}$$

The deviatoric stress can then be further split as

$$\boldsymbol{\tau} = \boldsymbol{\tau}_e + \boldsymbol{\tau}_v, \tag{68}$$

with $\boldsymbol{\tau}_e = 2\mu\,\mathbf{d}$ the elastic part and $\boldsymbol{\tau}_v = \boldsymbol{\tau} - \boldsymbol{\tau}_e$ the viscous. Such a decomposition is present within all other time-domain approaches to viscoelasticity with the GIA literature (e.g. Hanyk *et al.* 1995; Zhong *et al.* 2003; Latychev *et al.* 2005b; Bailey 2006). Applying the same decomposition to the stress within the adjoint problem, we can then write eq. (61) as

$$D_\eta \hat{J} = -\int_{\mathcal{I}} \frac{1}{2\mu\eta}\boldsymbol{\tau} : \left(\mathcal{T}\boldsymbol{\tau}_v^\dagger\right)\,\mathrm{d}t. \tag{69}$$

In this way, we see that the calculation of sensitivity kernels using the adjoint method does not depend on the use of internal variables for modelling the Maxwell rheology. Rather, any numerical code for modelling GIA is, subject to the minor modifications to the force terms discussed above, capable of determining all variables needed to form the sensitivity kernels.

3.6 Comparison with related studies

The adjoint equations and sensitivity kernels derived within this section are new, but they build on and extend results from earlier studies (e.g. Al-Attar & Tromp 2014; Martinec *et al.* 2015; Crawford *et al.* 2018; Al-Attar *et al.* 2024). In particular, Crawford *et al.* (2018) derived adjoint equations and sensitivity kernels for GIA within a Maxwell earth model in the absence of rotational feedbacks. Crawford's results were numerically implemented, with the forward calculations benchmarked against another GIA code, and tests performed to verify that the sensitivity kernels correctly predicted the linearized dependence of chosen objective functionals on the underlying model parameters. Subsequent work by Crawford (2019) and Lloyd *et al.* (2024) extended these numerical tests further by considering sensitivity kernels calculated relative to laterally heterogeneous viscosity models.

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If terms associated with rotational feedbacks are removed, then the results of this paper should be equivalent to those within Crawford *et al.* (2018). Verifying this correspondence is complicated due to the two papers being based on different, but equivalent, formulations of the GIA forward problem. Moreover, the adjoint variables in each paper are defined relative to their respective forward problems. This means, for example, that the adjoint sea level in this paper is not equal to the adjoint sea level within Crawford *et al.* (2018). Nevertheless, through a lengthy but simple algebraic process the required equivalence of the results can be established.

Beyond the direct calculation just mentioned, the consistency of our results with those of Crawford *et al.* (2018) can be seen in a number of ways. We noted in Section 3.3 that if shoreline migration is neglected, then the forward and adjoint problems have the same form except for the adjoint equations involving more general force terms. Precisely the same result was found to hold within Crawford *et al.* (2018) relative to their formulation of the forward GIA problem. Similarly, when shoreline migration is included within either approach, the adjoint problems involve the time-reversed ocean function obtained through solution of the forward problem.

A further useful point of comparison is with Al-Attar *et al.* (2024). That paper was concerned with the calculation of sensitivity kernels for the elastic sea level fingerprint problem incorporating rotational feedbacks, and contains detailed numerical checks of the theoretical results. The elastic fingerprint problem is a special case of the GIA problem in which viscoelasticity and shoreline migration are neglected. On making these approximations, it is readily seen that the results of this paper reduce precisely to those in Al-Attar *et al.* (2024).

3.7 Singular solutions of the adjoint equations

Within this section, we show that solutions of the adjoint equations possess singularities as functions of time whenever the objective functional depends on the state variables at a discrete set of times. Such behaviour cannot be dismissed as a mere pathology, but occurs frequently in practice. Suppose for simplicity that the objective functional, J, depends on the state variables at a single observation time, t'. The functional derivatives of J then take the form

$$D_{SL}J = \tilde{h}^{\dagger}_{SL}(\mathbf{x})\,\delta(t-t'), \tag{70}$$

$$D_{\mathbf{u}}J = \tilde{\mathbf{h}}^{\dagger}_{\mathbf{u}}(\mathbf{x})\,\delta(t-t'), \tag{71}$$

$$D_{\phi}J = \tilde{h}^{\dagger}_{\phi}(\mathbf{x})\,\delta(t-t'), \tag{72}$$

$$D_{\boldsymbol{\omega}}J = \tilde{\mathbf{h}}^{\dagger}_{\boldsymbol{\omega}}\,\delta(t-t'), \tag{73}$$

for some $(\tilde{h}^{\dagger}_{SL}, \tilde{\mathbf{h}}^{\dagger}_{\mathbf{u}}, \tilde{h}^{\dagger}_{\phi}, \tilde{\mathbf{h}}^{\dagger}_{\boldsymbol{\omega}})$. To see this, note that, for example we have

$$\int_{\mathcal{I}}\int_{\partial M} D_{SL}J\,\delta SL\,\mathrm{d}S\,\mathrm{d}t = \int_{\partial M}\tilde{h}^{\dagger}_{SL}(\mathbf{x})\,\delta SL(\mathbf{x},t')\,\mathrm{d}S, \tag{74}$$

with the term on the right-hand side depending on the sea level perturbation at the required observation time. Prior to the reversed observation time, $t'' = t_1 - t' + t_0$, it is clear that all adjoint state variables vanish. If we integrate the adjoint equations over the interval $(t''-\epsilon, t''+\epsilon)$ and take the limit $\epsilon \to 0$ we obtain

$$\mathcal{A}(\tilde{\mathbf{u}}^{\dagger},\tilde{\phi}^{\dagger}\,|\,\delta\mathbf{u},\delta\phi) + \int_{\partial M}(\delta\mathbf{u}\cdot\nabla\Phi + \delta\phi + \delta\psi)\,\tilde{\sigma}^{\dagger}\,\mathrm{d}S + \int_{\partial M}(\tilde{\mathbf{h}}^{\dagger}_{\mathbf{u}}\cdot\delta\mathbf{u} + \tilde{h}^{\dagger}_{\phi}\,\delta\phi)\,\mathrm{d}S$$
$$+ \int_{M}\rho\,(\tilde{\mathbf{u}}^{\dagger}\cdot\nabla\delta\psi + \delta\mathbf{u}\cdot\nabla\tilde{\psi}^{\dagger})\,\mathrm{d}^3\mathbf{x} + \delta\boldsymbol{\omega}\cdot\left(\mathbf{D}\cdot\tilde{\boldsymbol{\omega}}^{\dagger} + \tilde{\mathbf{h}}^{\dagger}_{\boldsymbol{\omega}}\right) + \int_{M}2\mu\left(\tau\left[\mathbf{m}^{\dagger}\right]^{+}_{-} - \tilde{\mathbf{d}}^{\dagger}\right) : \delta\mathbf{m}\,\mathrm{d}^3\mathbf{x} = 0, \tag{75}$$

$$\tilde{SL}^{\dagger}_{1} = -\frac{1}{g}(\tilde{\mathbf{u}}^{\dagger}\cdot\nabla\Phi + \tilde{\phi}^{\dagger} + \tilde{\psi}^{\dagger}) + \frac{\tilde{\Phi}^{\dagger}_{g}}{g}, \tag{76}$$

$$\tilde{\sigma}^{\dagger} = \rho_w\,C^{\dagger}\,\tilde{SL}^{\dagger}_{1} - \frac{1}{g}\tilde{h}^{\dagger}_{SL}, \tag{77}$$

$$\int_{\partial M}\tilde{\sigma}^{\dagger}\,\mathrm{d}S = 0, \tag{78}$$

where for a time-dependent variable, f, we have introduced notations

$$\tilde{f} = \lim_{\epsilon\to 0}\int_{t''-\epsilon}^{t''+\epsilon} f\,\mathrm{d}t, \quad [f]^{+}_{-} = \lim_{\epsilon\to 0}[f(t''+\epsilon) - f(t''-\epsilon)], \tag{79}$$

noting that $\tilde{f}$ vanishes for an integrable function. It follows that, with the exception of $\mathbf{m}^{\dagger}$, the adjoint state variables have delta-function singularities at t'', with their relative amplitudes determinable through solution of a *generalized fingerprint problem* of the form discussed in Al-Attar *et al.* (2024). Having done this, the finite-jump in the adjoint internal variable is given by

$$\left[\mathbf{m}^{\dagger}\right]^{+}_{-} = \frac{1}{\tau}\tilde{\mathbf{d}}^{\dagger}. \tag{80}$$

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For times $t > t''$, we can use $\left[\mathbf{m}^\dagger\right]^+_-$ as an initial condition to integrate the evolution equation for $\mathbf{m}^\dagger$, obtaining smooth solutions for all adjoint variables within the interval $(t'', t_1]$. By linearity of the adjoint equations, these arguments extend trivially to cases where the objective functional depends on the state variables at a finite number of observation times.

These results show that between observation times the adjoint GIA problem can be time stepped using standard methods. At each observation time, however, it is necessary to solve a generalized fingerprint problem whose solution determines a finite-jump in $\mathbf{m}^\dagger$ along with singular contributions to the other adjoint state variables. For the viscosity kernel in eq. (61), all terms within the integrand are bounded and hence the integral can be evaluated using a standard quadrature scheme applied within each subinterval between the observation times. In the case of the initial sea level kernel in eq. (64), both $D_{SL}J$ and the adjoint sea level lead to delta-function singularities at the observation times. Using the defining property of the delta-function, each singularity makes a discrete contribution to the integral whose value is determined through solution of the generalized fingerprint problem, while between the observation times the integrand is smooth and a standard quadrature scheme can be applied.

3.8 Singular ice kernel and gradient-based optimization

Singularities within the solution of the adjoint GIA problem have no effect on the use of sensitivity kernels for viscosity and initial sea level, and complicate their calculation only slightly. In the case of the ice thickness kernel,

$$D_I\hat{J} = -\rho_i\, g\left[(1-C)\left(\mathcal{T}SL_1^\dagger\right) - (1-C_0)\int_{\mathcal{I}}\mathcal{T}SL_1^\dagger\,\mathrm{d}t\,\delta(t-t_0)\right], \tag{81}$$

the situation is more involved, with this kernel having delta-function singularities at the initial time and each observation time. Within the context of gradient-based optimization, it is typical for descent directions to be formed from linear combinations of the sensitivity kernel at the current and previous iterations. But we clearly cannot add these singular ice kernels to an ice thickness model. Before outlining methods for addressing this problem, it is worth emphasising that the singularities of the ice kernel are real features reflecting the physics of GIA. Within this quasi-static theory, the instantaneous application of a load at time, t', produces an instantaneous elastic response, and hence a discontinuous solution of the forward GIA problem. It follows that the linearized dependence of the state variables at time, t', can be decomposed into the sum of two terms, the first associated with elastic deformation at the observation time, and the second due to viscoelastic relaxation from earlier loading. It is the former contribution that produces the delta-function singularity within the ice thickness kernel.

The simplest method for working with the singular ice kernel is to parametrize the ice thickness using a finite dimensional set of continuous basis functions. We then need only consider the projection of the ice kernel along each basis function, this requiring integration of the singular kernel against continuous functions. The introduction of such a model parametrization is, however, necessarily ad hoc. Moreover, as the size of the basis set is increased, the projection of these singular kernels cannot converge point-wise, and hence non-physical ringing and other artefacts will be introduced.

A better method was presented by Al-Attar & Tromp (2014) and Crawford *et al.* (2018) using the idea of Backus (1970) for *quelling* singular kernels. In fact, it was this idea that motivated the introduction of the rate-formulation of the GIA problem, with the quelling process there being built directly into the description of the forward problem. The idea can be applied more generally, however. The key step is to not regard the ice thickness as a model parameter, but to define it implicitly through its initial value, I_0, and its time-derivative, $\dot{I}$. To proceed, we recall that the first-order change in $\hat{J}$ is given by

$$\delta\hat{J} = -\int_{\mathcal{I}}\int_{\partial M}\rho_i g\left[(1-C)\delta I - (1-C_0)\delta I_0\right]\left(\mathcal{T}SL_1^\dagger\right)\,\mathrm{d}S\,\mathrm{d}t, \tag{82}$$

and seek new kernels, $D_{I_0}\hat{J}$ and $D_{\dot{I}}\hat{J}$, such that we can equivalently write

$$\delta\hat{J} = \int_{\partial M} D_{I_0}\hat{J}\,\delta I_0\,\mathrm{d}S + \int_{\mathcal{I}}\int_{\partial M} D_{\dot{I}}\hat{J}\,\delta\dot{I}\,\mathrm{d}S\,\mathrm{d}t. \tag{83}$$

Requiring equality for arbitrary δI, integrating by parts, and solving a trivial ODE (ordinary differential equation), we find

$$D_{\dot{I}}\hat{J}(\mathbf{x},t) = -\rho_i g\int_t^{t_1}\left[1-C(\mathbf{x},s)\right]SL_1^\dagger(\mathbf{x},t_1-s+t_0)\,\mathrm{d}s, \quad D_{I_0}\hat{J}(\mathbf{x}) = \rho_i g\int_{\mathcal{I}}\left[C(\mathbf{x},t)-C_0(\mathbf{x})\right]SL_1^\dagger(\mathbf{x},t_1-t+t_0)\,\mathrm{d}t. \tag{84}$$

By inspection, $D_{\dot{I}}\hat{J}$ undergoes only finite jumps at the observation times, and hence it is sufficiently regular for use within standard gradient-based optimization schemes. Descent directions for I_0 and $\dot{I}$ can, therefore, be obtained in a normal manner, and through a further time-integration they define an update for the ice thickness that is continuous in time.

A third, and likely best, approach is suggested by the recent work of Zuberi & Pratt (2017) and Syvret (2022) in seismic tomography. The application of this method to GIA will be discussed in detail elsewhere. For the moment, we just note that it has a rigorous foundation in functional analysis based on the Sobolev embedding theorem (e.g. Trèves 1975), and allows for stronger regularity requirements to be imposed on the ice thickness (and other model parameters) in both time and space.

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3.9 Constraining the initial sea level

Within the GIA forward problem, the initial sea level has been regarded as a parameter. In practice, however, this value is usually constrained by requiring that the calculated present-day sea level matches the observed value. Let $t_\mathrm{p} \leq t_1$ denote the time of the present and SL_p the observed sea level. For a given initial sea level, SL_0, we can compute $SL(t_\mathrm{p})$ through solution of the GIA forward problem. In general, this will not agree with SL_p, but a simple iterative procedure can be applied that converges rapidly (e.g. Johnston 1993; Peltier 1994; Mitrovica & Milne 2003; Kendall *et al.* 2005). Let SL_0^i be the ith estimate of initial sea level and $SL^i(t_\mathrm{p})$ the corresponding present-day prediction. We then update the initial sea level by setting

$$SL_0^{i+1} = SL_0^i - [SL^i(t_\mathrm{p}) - SL_\mathrm{p}]. \tag{85}$$

An alternative approach for determining the initial sea level has been recently discussed by Crawford (2019) and Lloyd *et al.* (2024). Within this method, the objective functional

$$J = \frac{1}{2}\int_{\partial M} [SL(t_\mathrm{p}) - SL_\mathrm{p}]^2 \,\mathrm{d}S, \tag{86}$$

is minimized with respect to SL_0 using gradient-based optimization. Applying the results of this paper, we can see that the initial sea level kernel for this objective functional takes the form

$$D_{SL_0}\hat{J} = [SL(t_\mathrm{p}) - SL_\mathrm{p}] - \rho_w g \int_{\mathcal{I}} (C - C_0)\left(\mathcal{T}SL_1^\dagger\right)\,\mathrm{d}t, \tag{87}$$

where $\mathrm{SL}_1^\dagger$ is obtained by solving the adjoint GIA problem subject to

$$D_{SL}J = [SL(t_\mathrm{p}) - SL_\mathrm{p}]\,\delta(t - t_\mathrm{p}), \tag{88}$$

defining the only non-zero force. If shoreline migration is neglected, then the second term in $D_{SL_0}\hat{J}$ vanishes, and this kernel is equal to the negative of the update to the initial sea level within eq. (85). It follows that the traditional iterative approach for matching present-day sea level can be viewed as an approximation to the steepest-descent minimization of eq. (86).

The constraint of matching the present-day sea level can be built directly into the definition of the GIA forward problem. It is then a relatively simple matter to reformulate the adjoint problem. We first write down a modified Lagrangian

$$\begin{aligned}
J + \int_{\mathcal{I}} \Bigg\{ & \mathcal{A}(\mathbf{u}, \phi \,|\, \mathcal{T}\mathbf{u}^\dagger, \mathcal{T}\phi^\dagger) - \int_M 2\mu \mathbf{m} : (\mathcal{T}\mathbf{d}^\dagger)\,\mathrm{d}^3\mathbf{x} + \int_{\partial M} \left[(\mathcal{T}\mathbf{u}^\dagger)\cdot\nabla\Phi + \mathcal{T}\phi^\dagger + \mathcal{T}\psi^\dagger\right]\sigma\,\mathrm{d}S \\
& + \int_M \rho\left[(\mathcal{T}\mathbf{u}^\dagger)\cdot\nabla\psi + \mathbf{u}\cdot\nabla(\mathcal{T}\psi^\dagger)\right]\mathrm{d}^3\mathbf{x} + (\mathcal{T}\boldsymbol{\omega}^\dagger)\cdot\mathbf{D}\cdot\boldsymbol{\omega} + \int_M 2\mu\,(\tau\,\dot{\mathbf{m}} + \mathbf{m} - \mathbf{d}) : (\mathcal{T}\mathbf{m}^\dagger)\,\mathrm{d}^3\mathbf{x} \\
& + \int_{\partial M} [\sigma - \rho_w\, C\, SL - \rho_i\,(1 - C)\, I + \rho_w\, C_0\, SL_0 + \rho_i\,(1 - C_0)\, I_0]\, g\,\left(\mathcal{T}SL_1^\dagger\right)\,\mathrm{d}S \\
& + \int_{\partial M} \left(g\,SL_1 + \mathbf{u}\cdot\nabla\Phi + \phi + \psi - \Phi_g\right)(\mathcal{T}\sigma^\dagger)\,\mathrm{d}S - (\mathcal{T}\Phi_g^\dagger)\int_{\partial M}\sigma\,\mathrm{d}S\Bigg\}\,\mathrm{d}t \\
& - \int_{\partial M} \rho_w g[SL(t_p) - SL_p]SL_0^\dagger\,\mathrm{d}S.
\end{aligned} \tag{89}$$

The only differences from the Lagrangian in eq. (46) are that SL_0 is here a state variable and the addition of the final term – with a corresponding Lagrange multiplier, $SL_0^\dagger$ – associated with the constraint $SL(t_\mathrm{p}) = SL_\mathrm{p}$. Derivation of the adjoint equations proceeds almost as before, with only the variation with respect to sea level being modified. To do this, it is useful to decompose the sea level as $SL = SL_0 + SL_1$ and vary the two terms separately. By letting L be stationary with respect to SL_1 we obtain

$$\sigma^\dagger = \rho_w C^\dagger SL_1^\dagger - \frac{1}{g}\mathcal{T}D_{SL}J + \rho_w SL_0^\dagger\,\delta(t_1 - t + t_0 - t_\mathrm{p}), \tag{90}$$

while by varying SL_0 we find

$$SL_0^\dagger = \int_{\mathcal{I}} (C^\dagger - C_0)SL_1^\dagger\,\mathrm{d}t - \frac{1}{g}\int_{\mathcal{I}} D_{SL}J\,\mathrm{d}t. \tag{91}$$

Combining the two results, we see that the adjoint load takes the form

$$\sigma^\dagger = \rho_w C^\dagger SL_1^\dagger - \frac{1}{g}\mathcal{T}D_{SL}J + \rho_w\left[\int_{\mathcal{I}} (C^\dagger - C_0)SL_1^\dagger\,\mathrm{d}t - \frac{1}{g}\int_{\mathcal{I}} D_{SL}J\,\mathrm{d}t\right]\delta(t_1 - t + t_0 - t_\mathrm{p}), \tag{92}$$

while all other adjoint equations are unchanged. The singular part of the adjoint load at the time-reversed present depends on the adjoint sea level over the whole time interval, and so these adjoint equations are not amenable to a direct time-integration. However, a simple iterative scheme could be applied, with an initial value for $SL_0^\dagger$ guessed and then successively refined. Expressions for the sensitivity kernels for viscosity and ice thickness are unchanged, but now they incorporate an implicit change in the initial sea level such that the calculated present-day sea level always matches SL_p.

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4 HESSIAN KERNELS AND THE SECOND-ORDER ADJOINT GIA PROBLEM

Within this section, we apply the second-order adjoint method to the GIA problem. These results extend earlier discussions within the literature that have considered only the first-order theory. The purpose of the second-order adjoint method is that it allows for second-derivatives of an objective functional to be calculated at a practicable cost. In particular, the action of the Hessian operator on a given model perturbation can be determined at a cost equivalent to four GIA simulations. Such calculations are necessary within the application of efficient Newton-type optimization schemes, and also within a range of methods for uncertainty quantification for non-linear inverse problems.

4.1 Hessian operators and kernels

Consider again an objective functional, J, and its reduced form, $\hat{J}$, defined in terms of the model parameters in the GIA forward problem. To simplify notations, we initially take viscosity, η, to be the only model parameter. For a given viscosity perturbation, the second-order functional derivative of $\hat{J}$ can be defined through the following Taylor expansion

$$\hat{J}(\eta+\delta\eta) = \hat{J}(\eta) + \int_M D_\eta \hat{J}(\mathbf{x})\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x} + \frac{1}{2}\int_M\int_M D^2_{\eta\eta}\hat{J}(\mathbf{x},\mathbf{x}')\,\delta\eta(\mathbf{x}')\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x}'\,\mathrm{d}^3\mathbf{x} + \cdots, \tag{93}$$

where spatial arguments have been included for clarity. Equivalently, the second-order functional derivative can be defined through the following limit

$$\int_M\int_M D^2_{\eta\eta}\hat{J}(\mathbf{x},\mathbf{x}')\,\Delta\eta(\mathbf{x}')\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x}'\,\mathrm{d}^3\mathbf{x} = \lim_{r\to 0}\lim_{s\to 0}\frac{1}{rs}\Big[\hat{J}(\eta + r\,\delta\eta + s\,\Delta\eta) - \hat{J}(\eta)\Big], \tag{94}$$

where $\delta\eta$ and $\Delta\eta$ are arbitrary viscosity perturbations. This latter definition makes it clear that $D^2_{\eta\eta}\hat{J}$ is symmetric in its spatial arguments. If we define a new functional

$$\hat{J}_2(\eta) = \lim_{s\to 0}\frac{1}{s}\Big[\hat{J}(\eta + s\,\Delta\eta) - \hat{J}(\eta)\Big] = \int_M D_\eta\hat{J}\,\Delta\eta\,\mathrm{d}^3\mathbf{x}, \tag{95}$$

then we have

$$\int_M D_\eta\hat{J}_2\,\delta\eta\,\mathrm{d}^3\mathbf{x}' = \lim_{r\to 0}\frac{1}{r}\Big[\hat{J}_2(\eta + r\,\delta\eta) - \hat{J}_2(\eta)\Big] = \int_M\int_M D^2_{\eta\eta}\hat{J}(\mathbf{x},\mathbf{x}')\,\Delta\eta(\mathbf{x}')\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x}'\,\mathrm{d}^3\mathbf{x}, \tag{96}$$

which implies

$$D_\eta\hat{J}_2(\mathbf{x}) = \int_M D^2_{\eta\eta}\hat{J}(\mathbf{x},\mathbf{x}')\,\Delta\eta(\mathbf{x}')\,\mathrm{d}^3\mathbf{x}'. \tag{97}$$

The *Hessian operator*, $\hat{\mathcal{H}}_{\eta\eta}$, of $\hat{J}$ at η acts on a viscosity perturbation, $\Delta\eta$, through

$$(\hat{\mathcal{H}}_{\eta\eta}\,\Delta\eta)(\mathbf{x}) = \int_M D^2_{\eta\eta}\hat{J}(\mathbf{x},\mathbf{x}')\,\Delta\eta(\mathbf{x}')\,\mathrm{d}^3\mathbf{x}', \tag{98}$$

with the result lying in the space of viscosity perturbations. Due to the symmetry of $D^2_{\eta\eta}\hat{J}$, we see that $\hat{\mathcal{H}}_{\eta\eta}$ is self-adjoint. Given this definition, the second-order Taylor expansion of $\hat{J}$ takes the form

$$\hat{J}(\eta+\delta\eta) = \hat{J}(\eta) + \int_M D_\eta\hat{J}(\mathbf{x})\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x} + \frac{1}{2}\int_M(\hat{\mathcal{H}}_{\eta\eta}\,\delta\eta)(\mathbf{x})\,\delta\eta(\mathbf{x})\,\mathrm{d}^3\mathbf{x} + \cdots, \tag{99}$$

where the second-order term is expressed as an integral of $\delta\eta$ against the function $\hat{\mathcal{H}}_{\eta\eta}\,\delta\eta$; we call this latter term the *Hessian kernel* of $\hat{J}$ relative to $\delta\eta$ (c.f. Fichtner & Trampert 2011). From eq. (97), we see that the Hessian kernel relative to $\Delta\eta$ is equal to the functional derivative of $\hat{J}_2$, with this identity being key to our discussion of second-order adjoint methods below.

These definitions extend readily to allow for the other model parameters within the GIA forward problem. The full Hessian operator has a block structure

$$\hat{\mathcal{H}} = \begin{pmatrix} \hat{\mathcal{H}}_{\eta\eta} & \hat{\mathcal{H}}_{\eta I} & \hat{\mathcal{H}}_{\eta SL_0} \\ \hat{\mathcal{H}}_{I\eta} & \hat{\mathcal{H}}_{II} & \hat{\mathcal{H}}_{ISL_0} \\ \hat{\mathcal{H}}_{SL_0\eta} & \hat{\mathcal{H}}_{SL_0 I} & \hat{\mathcal{H}}_{SL_0SL_0} \end{pmatrix}. \tag{100}$$

Here, for example, $\hat{\mathcal{H}}_{I\eta}$ is a linear operator mapping viscosity perturbations into ice thickness perturbations. Again, $\hat{\mathcal{H}}$ is self-adjoint, this implying identities such as $\hat{\mathcal{H}}_{\eta I} = \hat{\mathcal{H}}^*_{I\eta}$ with superscript, $*$, denoting the operator adjoint. The Hessian kernels (one for each model parameter) relative to a given model perturbation, $(\Delta\eta, \Delta I, \Delta SL_0)$, are defined in the obvious manner, and can be identified with the functional derivatives of

$$\hat{J}_2 = \int_M D_\eta\hat{J}\,\Delta\eta\,\mathrm{d}^3\mathbf{x} + \int_{\mathcal{I}}\int_{\partial M} D_I\hat{J}\,\Delta I\,\mathrm{d}S\,\mathrm{d}t + \int_{\partial M} D_{SL_0}\hat{J}\,\Delta SL_0\,\mathrm{d}S. \tag{101}$$

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4.2 Second-order PDE constrained optimization through the Lagrangian method

To derive the second-order adjoint equations, we follow the method of Syvret (2022). With L the Lagrangian in eq. (46), and for fixed model perturbations, $(\Delta\eta,\, \Delta I,\, \Delta SL_0)$, we can define a new functional

$$J_2 = \int_M D_\eta L\, \Delta\eta\, \mathrm{d}^3\mathbf{x} + \int_{\mathcal{I}}\int_{\partial M} D_I L\, \Delta I\, \mathrm{d}S\, \mathrm{d}t + \int_{\partial M} D_{SL_0} L\, \Delta SL_0\, \mathrm{d}S, \tag{102}$$

depending on the state variables $(\mathbf{u},\, \phi,\, \boldsymbol{\omega},\, \mathbf{m},\, SL,\, \sigma,\, \Phi_g)$, the adjoint state variables $(\mathbf{u}^\dagger,\, \phi^\dagger,\, \boldsymbol{\omega}^\dagger,\, \mathbf{m}^\dagger,\, SL^\dagger,\, \sigma^\dagger,\, \Phi_g^\dagger)$, and the model parameters $(\eta,\, I,\, SL_0)$. Using the results of the first-order adjoint theory, this functional can be written explicitly as

$$\begin{aligned} J_2 = &\int_{\mathcal{I}}\int_M \frac{2\mu}{\eta}(\mathbf{m}-\mathbf{d}) : (\mathcal{T}\mathbf{m}^\dagger)\, \Delta\eta\, \mathrm{d}^3\mathbf{x}\, \mathrm{d}t - \int_{\mathcal{I}}\int_{\partial M} \rho_i g\, [(1-C)\, \Delta I - (1-C_0)\, \Delta I_0]\, (\mathcal{T}SL_1^\dagger)\, \mathrm{d}S\, \mathrm{d}t \\ &+ \int_{\mathcal{I}}\int_{\partial M} \left[D_{SL} J - \rho_w g\, (C-C_0)\left(\mathcal{T}SL_1^\dagger\right)\right] \Delta SL_0\, \mathrm{d}S\, \mathrm{d}t. \end{aligned} \tag{103}$$

By requiring that the arguments of J_2 satisfy both the forward and first-order adjoint GIA problems, we have the equality $J_2 = \hat{J}_2$ with $\hat{J}_2$ defined in eq. (101). It follows that the Hessian kernels of $\hat{J}$ relative to the chosen model perturbations can be obtained by differentiating J_2 subject to these constraints. To proceed, we define a new Lagrangian

$$\begin{aligned} L_2 = J_2 &+ \int_{\mathcal{I}} \Bigg\{ \mathcal{A}(\mathbf{u},\, \phi \,|\, \mathcal{T}\Delta\mathbf{u}^\dagger,\, \mathcal{T}\Delta\phi^\dagger) - \int_M 2\mu\mathbf{m} : (\mathcal{T}\Delta\mathbf{d}^\dagger)\, \mathrm{d}^3\mathbf{x} + \int_{\partial M} \left[(\mathcal{T}\Delta\mathbf{u}^\dagger)\cdot\nabla\Phi + \mathcal{T}\Delta\phi^\dagger + \mathcal{T}\Delta\psi^\dagger\right] \sigma\, \mathrm{d}S \\ &+ \int_M \rho\left[(\mathcal{T}\Delta\mathbf{u}^\dagger)\cdot\nabla\psi + \mathbf{u}\cdot\nabla(\mathcal{T}\Delta\psi^\dagger)\right] \mathrm{d}^3\mathbf{x} + (\mathcal{T}\Delta\boldsymbol{\omega}^\dagger)\cdot\mathbf{D}\cdot\boldsymbol{\omega} + \int_M 2\mu\,(\tau\,\dot{\mathbf{m}} + \mathbf{m} - \mathbf{d}) : (\mathcal{T}\Delta\mathbf{m}^\dagger)\, \mathrm{d}^3\mathbf{x} \\ &+ \int_{\partial M} [\sigma - \rho_w\, C\, SL - \rho_i\,(1-C)\, I + \rho_w\, C_0\, SL_0 + \rho_i\,(1-C_0)\, I_0]\, g\, (\mathcal{T}\Delta SL_1^\dagger)\, \mathrm{d}S \\ &+ \int_{\partial M} \left(g\, SL_1 + \mathbf{u}\cdot\nabla\Phi + \phi + \psi - \Phi_g\right)(\mathcal{T}\Delta\sigma^\dagger)\, \mathrm{d}S - (\mathcal{T}\Delta\Phi_g^\dagger)\int_{\partial M}\sigma\, \mathrm{d}S \Bigg\}\, \mathrm{d}t \\ &+ \mathcal{A}(\mathbf{u}^\dagger,\, \phi^\dagger \,|\, \mathcal{T}\Delta\mathbf{u},\, \mathcal{T}\Delta\phi) - \int_M 2\mu\, \mathbf{m}^\dagger : (\mathcal{T}\Delta\mathbf{d})\, \mathrm{d}^3\mathbf{x} + \int_{\partial M} [(\mathcal{T}\Delta\mathbf{u})\cdot\nabla\Phi + \mathcal{T}\Delta\phi + \mathcal{T}\Delta\psi]\, \sigma^\dagger\, \mathrm{d}S \\ &+ \int_{\partial M} \left[(\mathcal{T}D_{\mathbf{u}}J)\cdot(\mathcal{T}\Delta\mathbf{u}) + \left(\mathcal{T}D_\phi J\right)\,(\mathcal{T}\Delta\phi)\right] \mathrm{d}S + \int_M \rho\, \left[\mathbf{u}^\dagger\cdot\nabla(\mathcal{T}\Delta\psi) + (\mathcal{T}\Delta\mathbf{u})\cdot\nabla\psi^\dagger\right] \mathrm{d}^3\mathbf{x} \\ &+ (\mathcal{T}\Delta\boldsymbol{\omega})\cdot\left[\mathbf{D}\cdot\boldsymbol{\omega}^\dagger + (\mathcal{T}D_\omega J)\right] + \int_M 2\mu\left(\tau\dot{\mathbf{m}}^\dagger + \mathbf{m}^\dagger - \mathbf{d}^\dagger\right) : (\mathcal{T}\Delta\mathbf{m})\, \mathrm{d}^3\mathbf{x} \\ &+ \int_{\partial M}\left[\sigma^\dagger - \rho_w\, C^\dagger\, SL_1^\dagger + \frac{1}{g}\mathcal{T}D_{SL}J\right](g\mathcal{T}\Delta SL_1)\, \mathrm{d}S \\ &+ \int_{\partial M}\left(g\, SL_1^\dagger + \mathbf{u}^\dagger\cdot\nabla\Phi + \phi^\dagger + \psi^\dagger - \Phi_g^\dagger\right)(\mathcal{T}\Delta\sigma)\, \mathrm{d}S - (\mathcal{T}\Delta\Phi_g)\int_{\partial M}\sigma^\dagger\, \mathrm{d}S \Bigg\}\, \mathrm{d}t. \end{aligned} \tag{104}$$

Here, the *second-order adjoint state variables* $(\Delta\mathbf{u}^\dagger,\, \Delta\phi^\dagger,\, \Delta\boldsymbol{\omega}^\dagger,\, \Delta\mathbf{m}^\dagger,\, \Delta SL_1^\dagger,\, \Delta\sigma^\dagger,\, \Delta\Phi_g^\dagger)$ are associated with the constraint that the state variables solve the forward GIA problem, while the *second-order state variables* $(\Delta\mathbf{u},\, \Delta\phi,\, \Delta\boldsymbol{\omega},\, \Delta\mathbf{m},\, \Delta SL_1,\, \Delta\sigma,\, \Delta\Phi_g)$ do the same but for the first-order adjoint problem. Note that both sets of new variables have been time-reversed for convenience. The Lagrange multiplier theorem tells us that the functional derivatives of L_2 and $\hat{J}_2$ with respect to the model parameters coincide so long as L_2 is stationary with respect to its other variables. Varying the second-order state and adjoint state variables gives the forward and first-order adjoint problems. Requiring that L_2 is stationary with respect to the state and first-order adjoint state variables then gives two new sets of equations that the second-order state and adjoint state variables must satisfy.

By varying L_2 with respect to the first-order adjoint state variables we obtain

$$\begin{aligned} &\mathcal{A}(\Delta\mathbf{u},\, \Delta\phi \,|\, \delta\mathbf{m}^\dagger,\, \delta\phi^\dagger) - \int_M 2\mu\Delta\mathbf{m} : \delta\mathbf{d}^\dagger\, \mathrm{d}^3\mathbf{x} + \int_{\partial M}\left(\delta\mathbf{u}^\dagger\cdot\nabla\Phi + \delta\phi^\dagger + \delta\psi^\dagger\right)\Delta\sigma\, \mathrm{d}S \\ &+ \int_M \rho\left(\delta\mathbf{u}^\dagger\cdot\nabla\Delta\psi + \Delta\mathbf{u}\cdot\nabla\delta\psi^\dagger\right)\, \mathrm{d}^3\mathbf{x} + \delta\boldsymbol{\omega}^\dagger\cdot\mathbf{D}\cdot\Delta\boldsymbol{\omega} + \int_M 2\mu\left[\tau\Delta\dot{\mathbf{m}} + \Delta\mathbf{m} - \Delta\mathbf{d} + \frac{\Delta\eta}{\eta}(\mathbf{m}-\mathbf{d})\right] : \delta\mathbf{m}^\dagger\, \mathrm{d}^3\mathbf{x} = 0, \end{aligned} \tag{105}$$

$$\Delta\sigma = \rho_w C\Delta SL_1 + \rho_w(C-C_0)\,\Delta SL_0 + \rho_i\,[(1-C)\,\Delta I - (1-C_0)\,\Delta I_0], \tag{106}$$

$$\Delta SL_1 = -\frac{1}{g}\,(\Delta\mathbf{u}\cdot\nabla\Phi + \Delta\phi + \Delta\psi) + \frac{\Delta\Phi_g}{g}, \tag{107}$$

$$\int_{\partial M}\Delta\sigma\, \mathrm{d}S = 0. \tag{108}$$

where in the first equation $(\delta\mathbf{u}^\dagger,\, \delta\phi^\dagger,\, \delta\boldsymbol{\omega}^\dagger,\, \delta\mathbf{m}^\dagger)$ are arbitrary time-independent test functions, along with the initial condition, $\Delta\mathbf{m}(\mathbf{x},\, t_0) = \mathbf{0}$. These equations can be seen to be the linearization of the forward GIA problem with respect to the model parameters, this being a general

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feature of the second-order adjoint method. We note that the linearization of the GIA forward problem with respect to its model parameters has also been discussed by Martinec *et al.* (2015) in the context of what they term the "forward sensitivity method".

Calculating the variation of L_2 with respect to the state variables is more involved due to the dependence of the forces within the first-order adjoint equations on the state variables. For example, consider the term

$$\int_{\mathcal{I}}\int_{\partial M} (\mathcal{T} D_{\mathbf{u}} J)\cdot(\mathcal{T}\Delta\mathbf{u})\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t = \int_{\mathcal{I}}\int_{\partial M} D_{\mathbf{u}} J\cdot\Delta\mathbf{u}\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t, \tag{109}$$

with the equality following from the properties of $\mathcal{T}$. If we vary this functional with respect to $\mathbf{u}$, we obtain

$$\int_{\mathcal{I}}\int_{\partial M} (\mathcal{H}_{\mathbf{uu}}\Delta\mathbf{u})\cdot\delta\mathbf{u}\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t, \tag{110}$$

where we have recalled the definition of the Hessian operator of a functional; note that the Hessian of $\hat{J}$ has a hat on, but that for J does not. This idea extends readily to the variations of this functional with respect to other state variables, and to the other adjoint forces within L_2. Using these notations, the variation of L_2 with respect to the state variables yields the following set of equations

$$\begin{aligned}
&\mathcal{A}(\Delta\mathbf{u}^\dagger,\Delta\phi^\dagger\,|\,\delta\mathbf{m},\delta\phi) - \int_M 2\mu\left(\Delta\mathbf{m}^\dagger - \frac{\Delta\eta}{\eta}\mathbf{m}^\dagger\right) : \delta\mathbf{d}\,\mathrm{d}^3\mathbf{x} + \int_{\partial M}(\delta\mathbf{u}\cdot\nabla\Phi + \delta\phi + \delta\psi)\,\Delta\sigma^\dagger\,\mathrm{d}S\\
&+\int_M \rho\left(\delta\mathbf{u}\cdot\nabla\Delta\psi^\dagger + \Delta\mathbf{u}^\dagger\cdot\nabla\delta\psi\right)\mathrm{d}^3\mathbf{x} + \delta\boldsymbol{\omega}\cdot\mathbf{D}\cdot\Delta\boldsymbol{\omega}^\dagger + \int_M 2\mu\left[\tau\Delta\dot{\mathbf{m}}^\dagger + \Delta\mathbf{m}^\dagger - \Delta\mathbf{d}^\dagger + \frac{\Delta\eta}{\eta}\mathbf{m}^\dagger\right] : \delta\mathbf{m}\,\mathrm{d}^3\mathbf{x}\\
&+\int_{\partial M}\left(\mathcal{T}\mathcal{H}_{\mathbf{uu}}\,\Delta\mathbf{u} + \mathcal{T}\mathcal{H}_{\mathbf{u}\phi}\,\Delta\phi + \mathcal{T}\mathcal{H}_{\mathbf{u}\boldsymbol{\omega}}\,\Delta\boldsymbol{\omega}\right)\cdot\delta\mathbf{u}\,\mathrm{d}S + \int_{\partial M}\left(\mathcal{T}\mathcal{H}_{\phi\mathbf{u}}\,\Delta\mathbf{u} + \mathcal{T}\mathcal{H}_{\phi\phi}\,\Delta\phi + \mathcal{T}\mathcal{H}_{\phi\boldsymbol{\omega}}\,\Delta\boldsymbol{\omega}\right)\delta\phi\,\mathrm{d}S\\
&+\left(\mathcal{T}\mathcal{H}_{\boldsymbol{\omega}\mathbf{u}}\,\Delta\mathbf{u} + \mathcal{T}\mathcal{H}_{\boldsymbol{\omega}\phi}\,\Delta\phi + \mathcal{T}\mathcal{H}_{\boldsymbol{\omega}\boldsymbol{\omega}}\,\Delta\boldsymbol{\omega}\right)\cdot\delta\boldsymbol{\omega} = 0,
\end{aligned} \tag{111}$$

$$\begin{aligned}
\Delta\sigma^\dagger = {}& \rho_w C^\dagger\,\Delta SL_1^\dagger - \rho_w\mathcal{T}\left(\frac{\rho_w\Delta SL - \rho_i\Delta I}{\rho_w\partial_\perp SL - \rho_i\partial_\perp I}\delta_{\mathcal{O}_t}\right)SL_1^\dagger\\
&-\frac{1}{g}\mathcal{T}\left(\mathcal{H}_{SLSL}\Delta SL + \mathcal{H}_{SL\mathbf{u}}\Delta\mathbf{u} + \mathcal{H}_{SL\phi}\Delta\phi + \mathcal{H}_{SL\boldsymbol{\omega}}\Delta\boldsymbol{\omega}\right),
\end{aligned} \tag{112}$$

$$\Delta SL_1^\dagger = -\frac{1}{g}\left(\Delta\mathbf{u}^\dagger\cdot\nabla\Phi + \Delta\phi^\dagger + \Delta\psi^\dagger\right) + \frac{\Delta\Phi_g^\dagger}{g}, \tag{113}$$

$$\int_{\partial M}\Delta\sigma^\dagger\,\mathrm{d}S = 0, \tag{114}$$

where $(\delta\mathbf{d},\,\delta\phi,\,\delta\boldsymbol{\omega},\,\delta m)$ are arbitrary time-independent test functions, we have defined $\Delta SL = \Delta SL_0 + \Delta SL_1$, and we have the initial condition, $\Delta\mathbf{m}^\dagger(\mathbf{x},t_0) = \mathbf{0}$.

4.3 Practical implementation

Within the second-order adjoint theory there are four equations to be solved. First, there is the forward GIA problem. Next, we have the first-order adjoint equations which are linear but with coefficients and force terms dependent on the solution of the forward problem. Solution of these first two equations is, of course, necessary for the calculation of sensitivity kernels via the first-order adjoint method. The final two equations determine the second-order state and adjoint state variables, with both sets of equations depending explicitly on the assumed model perturbations $(\Delta\eta,\,\Delta I,\,\Delta SL_0)$. The equations for the second-order state variables depend on the solution of the forward problem due to the occurrence of the ocean function, C, but are independent of either set of adjoint variables. Finally, the second-order adjoint variables can be determined, with the force terms in this case depending on the solution of the state variables, first-order adjoint state variables, and second-order state variables. The sequence of dependences just described means that the four equations can be solved sequentially.

As with the first-order adjoint theory, the new equations to be solved resemble very closely the forward GIA problem, with the principal difference being the occurrence of additional force terms. Within the second-order equations the additional forces within the elastostatic subproblems include volumetric components associated with the given viscosity perturbation, $\Delta\eta$. Nevertheless, the modifications necessary to implement the second-order adjoint theory within existing GIA codes are modest.

The uniqueness and existence of solutions to the equations for the second-order state and adjoint state variables can be readily assessed. In particular, within the equations for the second-order adjoint state variables, the requirement that the additional force terms apply no net force nor torque to the earth model follows from the expansion of eq. (40) to second-order. Similarly, the second-order variables are defined only up to rigid body motions, but it will be seen shortly that such terms have no effect on the Hessian kernels.

Finally, we note that the equations for the second-order adjoint state variables contain force terms that can be singular in both space and time. As with the first-order adjoint equations, practical methods for dealing with these singularities can be developed but will be discussed in a future work.

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4.4 Expressions for the Hessian kernels

By differentiating the Lagrangian, L_2, with respect to the model parameters, we can write the functional derivatives of $\hat{J}_2$ in terms of the solution of the state and adjoint state variables. As noted previously, these functional derivatives are equal to the components of the action of the Hessian of $\hat{J}$ on the selected model perturbations. In the case of viscosity the calculations are easy, with the result

$$D_\eta \hat{J}_2 = \int_{\mathcal{I}} \frac{2\mu}{\eta} \left[(\mathbf{m} - \mathbf{d}) : \mathcal{T}\left(\Delta\mathbf{m}^\dagger - \frac{\Delta\eta}{\eta}\mathbf{m}^\dagger \right) + (\mathbf{m}^\dagger - \mathbf{d}^\dagger) : (\mathcal{T}\Delta\mathbf{m}) \right] \mathrm{d}t. \tag{115}$$

Functional derivatives with respect to ice thickness and initial sea level are more complicated due to the dependence of the ocean function on these fields, but following a now routine calculation using eq. (A30) of Crawford *et al.* (2018) we find

$$\begin{aligned} D_I \hat{J}_2 = &-\rho_i g \left[(1 - C)(\mathcal{T}\Delta SL_1^\dagger) - (1 - C_0) \int_{\mathcal{I}} \mathcal{T}\Delta SL_1^\dagger \, \mathrm{d}t \, \delta(t - t_0) \right] + \rho_i g \frac{\rho_w \Delta SL - \rho_i \Delta I}{\rho_w \partial_\perp SL - \rho_i \partial_\perp I} (\mathcal{T} SL_1^\dagger) \, \delta_{\partial\mathcal{O}_t} \\ &- \rho_i g \frac{\rho_w \Delta SL_0 - \rho_i \Delta I_0}{\rho_w \partial_\perp SL_0 - \rho_i \partial_\perp I_0} \int_{\mathcal{I}} \mathcal{T} SL_1^\dagger \, \mathrm{d}t \, \delta_{\partial\mathcal{O}_{t_0}} \, \delta(t - t_0), \end{aligned} \tag{116}$$

$$\begin{aligned} D_{SL_0} \hat{J}_2 = &\int_{\mathcal{I}} \left[\mathcal{H}_{SLSL} \Delta SL_0 - \rho_w g (C - C_0)(\mathcal{T}\Delta SL_1^\dagger) \right] \mathrm{d}t - \int_{\mathcal{I}} \rho_w g \frac{\rho_w \Delta SL - \rho_i \Delta I}{\rho_w \partial_\perp SL - \rho_i \partial_\perp I} \left(\mathcal{T} SL_1^\dagger \right) \delta_{\mathcal{O}_t} \, \mathrm{d}t \\ &+ \rho_w g \frac{\rho_w \Delta SL_0 - \rho_i \Delta I_0}{\rho_w \partial_\perp SL_0 - \rho_i \partial_\perp I_0} \int_{\mathcal{I}} \mathcal{T} SL_1^\dagger \, \mathrm{d}t \, \delta_{\mathcal{O}_{t_0}}, \end{aligned} \tag{117}$$

where we recall that $\Delta SL = \Delta SL_0 + \Delta SL_1$. By inspection, non-uniqueness within the solution of the various elastostatic subproblems does not propagate through to the Hessian kernels.

Looking at the above expressions, we note that certain terms within the Hessian kernels are independent of the second-order variables. This leads to the following approximate formulae:

$$D_\eta \hat{J}_2 \approx -\int_{\mathcal{I}} \frac{2\mu\Delta\eta}{\eta^2} (\mathbf{m} - \mathbf{d}) : (\mathcal{T}\mathbf{m}^\dagger) \, \mathrm{d}t, \tag{118}$$

$$D_I \hat{J}_2 \approx -\rho_i g \frac{\rho_w \Delta SL_0 - \rho_i \Delta I_0}{\rho_w \partial_\perp SL_0 - \rho_i \partial_\perp I_0} \int_{\mathcal{I}} \mathcal{T} SL_1^\dagger \, \mathrm{d}t \, \delta_{\partial\mathcal{O}_{t_0}} \, \delta(t - t_0), \tag{119}$$

$$D_{SL_0} \hat{J}_2 \approx \int_{\mathcal{I}} \mathcal{H}_{SLSL} \Delta SL_0 \, \mathrm{d}t + \rho_w g \frac{\rho_w \Delta SL_0 - \rho_i \Delta I_0}{\rho_w \partial_\perp SL_0 - \rho_i \partial_\perp I_0} \int_{\mathcal{I}} \mathcal{T} SL_1^\dagger \, \mathrm{d}t \, \delta_{\mathcal{O}_{t_0}}, \tag{120}$$

that could be used to approximate the action of the Hessian within the context of a quasi-Newton optimization scheme without having to solve for the second-order variables.

5 DISCUSSION

The aims of this paper have been three-fold. First, we have extended adjoint GIA theory to account for rotational feedbacks. This aspect of the problem was (knowingly) neglected within earlier discussions, but it is sufficiently important that it must be included in any serious practical application. The second was to present a version of the adjoint GIA equations that is independent of the rate-formulation used by Al-Attar & Tromp (2014) and Crawford *et al.* (2018). While the rate-formulation has its merits, its unfamiliarity has probably limited the adoption of adjoint methods within the GIA community. Moreover, the apparent need for explicit time stepping schemes severely limits its numerical efficiency. Finally, details of the second-order adjoint theory for the GIA problem have been documented in full. Within future work on the GIA inverse problem, a key issue to address is uncertainty within the generated models. Second-order adjoint methods underlie the few techniques available for uncertainty quantification within large-scale non-linear inverse problems, and hence we expect that there will be growing need for these results in the years to come.

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DATA AVAILABILITY

There is no data associated with this paper.

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APPENDIX A: ACCOUNTING FOR A FLUID CORE

A1 Geometry of the earth model

As in the main text, we let M denote the volume occupied by the equilibrium earth model. We now suppose that this set is decomposed as $M_S \cup M_F$ where M_S denotes solid regions and M_F fluid. The precise number of solid and fluid regions need not be specified, but we assume that the different regions are nested one within another, and that the outermost region is solid. We write Σ for the union of all internal and external boundaries, with this set decomposed as $\Sigma = \partial M \cup \Sigma_{FS} \cup \Sigma_{SF}$, where Σ_{FS} (resp. Σ_{SF}) denotes boundaries between fluid and solid regions where the fluid is on the inner (resp. outer) side of the boundary. Note that internal boundaries between solid regions are permitted, representing discontinuities in physical parameters, but such boundaries require no special consideration within the equations of motion and so will not be explicitly discussed.

A2 Elastostatic problem including fluid regions

To model elastostatic deformation in such an earth model, we follow Dahlen (1974) who showed that the linearized Lagrangian displacement cannot be well-defined within fluid regions, but that an Eulerian formulation can instead be used. Moreover, due to the hydrostatic equilibrium condition within fluid regions, all necessary dynamical fields can be expressed in terms of the gravitational potential perturbation. Bagheri *et al.* (2019) generalized Dahlen's arguments slightly to account for an applied tidal potential, showing that perturbations to density and pressure within fluid regions are given by

$$\rho_1 = g^{-1}\partial_r \rho\,(\phi + \psi), \qquad p_1 = -\rho\,(\phi + \psi). \tag{A1}$$

Using these results, eq. (13) is generalized to

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S + \int_{M_S} \rho\,\mathbf{u}' \cdot \nabla\psi \,\mathrm{d}^3\mathbf{x}$$
$$+ \int_{M_F} g^{-1}\partial_r \rho\,\psi\,\phi' \,\mathrm{d}^3\mathbf{x} + \int_{\Sigma_{FS}} \rho^-\psi\,\hat{\mathbf{n}} \cdot \mathbf{u}' \,\mathrm{d}S - \int_{\Sigma_{FS}} \rho^+\psi\,\hat{\mathbf{n}} \cdot \mathbf{u}' \,\mathrm{d}S = 0, \tag{A2}$$

where $\rho^{\pm}$ denotes the density evaluated on the upper (+) or lower (-) side of a boundary. Here, the bilinear form $\mathcal{A}$ is now given by

$$\begin{aligned}\mathcal{A}(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') = &\int_{M_S} \kappa\,\nabla\cdot\mathbf{u}\,\nabla\cdot\mathbf{u}'\,\mathrm{d}^3\mathbf{x} + \int_{M_S} 2\mu\,\mathbf{d}:\mathbf{d}'\,\mathrm{d}^3\mathbf{x} + \frac{1}{2}\int_{M_S}\rho\left[\nabla(\mathbf{u}\cdot\nabla\Phi)\cdot\mathbf{u}' + \nabla(\mathbf{u}'\cdot\nabla\Phi)\cdot\mathbf{u}\right]\mathrm{d}^3\mathbf{x}\\ &-\frac{1}{2}\int_{M_S}\rho\left(\nabla\cdot\mathbf{u}\,\nabla\Phi\cdot\mathbf{u}' + \nabla\cdot\mathbf{u}'\,\nabla\Phi\cdot\mathbf{u}\right)\mathrm{d}^3\mathbf{x} + \int_{M_S}\rho\left(\nabla\phi\cdot\mathbf{u}' + \nabla\phi'\cdot\mathbf{u}\right)\mathrm{d}^3\mathbf{x}\\ &+\frac{1}{4\pi G}\int_{\mathbf{R}^3}\nabla\phi\cdot\nabla\phi'\,\mathrm{d}^3\mathbf{x} + \int_{M_F} g^{-1}\partial_r\rho\,\phi\phi'\,\mathrm{d}^3\mathbf{x} + \int_{\Sigma_{FS}}\rho^{-}g\,\hat{\mathbf{n}}\cdot\mathbf{u}\,\hat{\mathbf{n}}\cdot\mathbf{u}'\,\mathrm{d}S\\ &-\int_{\Sigma_{SF}}\rho^{+}g\,\hat{\mathbf{n}}\cdot\mathbf{u}\,\hat{\mathbf{n}}\cdot\mathbf{u}'\,\mathrm{d}S + \int_{\Sigma_{FS}}\rho^{-}(\phi\mathbf{u}'+\phi'\mathbf{u})\cdot\hat{\mathbf{n}}\,\mathrm{d}S - \int_{\Sigma_{SF}}\rho^{+}(\phi\mathbf{u}'+\phi'\mathbf{u})\cdot\hat{\mathbf{n}}\,\mathrm{d}S,\end{aligned} \tag{A3}$$

as derived within Al-Attar & Tromp (2014). Notably, the symmetry of the bilinear form in eq. (6) is retained along with eq. (9) for linearized rigid body motions. It follows that our discussion of uniqueness and existence within the main paper remains valid.

A3 Rotational feedbacks

Our earlier discussion of rotational feedbacks followed Al-Attar *et al.* (2024), with the key identity being

$$\boldsymbol{\omega}'\cdot\mathbf{D}\cdot\boldsymbol{\omega} + \int_M \rho\,\mathbf{u}\cdot\nabla\psi'\,\mathrm{d}^3\mathbf{x} + \int_{\partial M}\sigma\,\psi'\,\mathrm{d}S = 0, \tag{A4}$$

which serves to define $\boldsymbol{\omega}$ in terms of the load and the associated displacement vector. Derivation of this result depends on the linearized relation between the displacement vector and the inertia tensor perturbation. Such a relation carries over to solid regions of the model, but within fluid regions a modified formula is required. To do this, we write the inertia tensor perturbation in fluid regions using Eulerian variables, with a volumetric contribution involving ρ_1 defined above, and boundary perturbations expressed in terms of the displacement vector on the solid-side. The result is the generalized relation

$$\boldsymbol{\omega}'\cdot\mathbf{D}\cdot\boldsymbol{\omega} + \int_{M_S}\rho\,\mathbf{u}\cdot\nabla\psi'\,\mathrm{d}^3\mathbf{x} + \int_{M_F} g^{-1}\partial_r\rho(\phi+\psi)\psi'\,\mathrm{d}^3\mathbf{x} + \int_{\Sigma_{FS}}\rho\psi'\mathbf{u}\cdot\hat{\mathbf{n}}\,\mathrm{d}S - \int_{\Sigma_{SF}}\rho\psi'\mathbf{u}\cdot\hat{\mathbf{n}}\,\mathrm{d}S + \int_{\partial M}\sigma\,\psi'\,\mathrm{d}S = 0, \tag{A5}$$

which is to hold for all $\boldsymbol{\omega}'$. This can be combined with the weak-form of the elastostatic problem to generalize eq. (17) to the case of fluid-solid earth models:

$$\begin{aligned}&\mathcal{A}(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') + \int_{\partial M}(\mathbf{u}'\cdot\nabla\Phi + \phi' + \psi')\,\sigma\,\mathrm{d}S + \int_{M_S}\rho\,\left(\mathbf{u}'\cdot\nabla\psi + \mathbf{u}\cdot\nabla\psi'\right)\mathrm{d}^3\mathbf{x} + \int_{M_F} g^{-1}\partial_r\rho\,\left(\psi\,\phi' + \psi'\,\phi + \psi\,\psi'\right)\mathrm{d}^3\mathbf{x}\\ &+\int_{\Sigma_{FS}}\rho^{-}\left(\psi\,\hat{\mathbf{n}}\cdot\mathbf{u}' + \psi'\,\hat{\mathbf{n}}\cdot\mathbf{u}\right)\mathrm{d}S - \int_{\Sigma_{FS}}\rho^{+}\left(\psi\,\hat{\mathbf{n}}\cdot\mathbf{u}' + \psi'\,\hat{\mathbf{n}}\cdot\mathbf{u}\right)\mathrm{d}S + \boldsymbol{\omega}'\cdot\mathbf{D}\cdot\boldsymbol{\omega} = 0.\end{aligned} \tag{A6}$$

The incorporation of viscoelasticity and sea level into the problem follows exactly as in the main text.

It would be easy, if tedious, to write down modified Lagrangians and carry through the full derivation of the first- and second-order adjoint problems, but this is not necessary. Instead, we need merely note that both eqs (17) and (A6) involve symmetric bilinear forms in the triplets $(\mathbf{u},\phi,\boldsymbol{\omega})$ and $(\mathbf{u}',\phi',\boldsymbol{\omega}')$. It is this symmetry alone that accounts for the elastostatic and rotational equations remaining unchanged between the forward and adjoint problems. Moreover, the model parameters of interest do not occur within the new terms linked to fluid regions, and hence expressions for the sensitivity and Hessian kernels in the main paper remain valid.

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